Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 26, 2026

1 Overview

This document compiles every current SCM case report TeX file under results/.

2 Case 1: cases99 80x2m 9h report

3 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/cases99/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

3.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/cases99`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

3.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	10.000000 $\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	0.000000 $\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	300.000000 $\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 $\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 $\mathrm{s^{-1}}$

3.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399988, 94.108850]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [-2.012 \times 10^{-6}, 1552.767352]$

3.4 Generated Artifacts

- Time-series CSV: `results/cases99/time_series.csv`
- Diagnostic summary JSON: `results/cases99/summary.json`
- Payload JLD2: `results/cases99/payload.jld2`

3.5 Figure Manifest

1. Time series of T__s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Closure manifold diagnostics (Km vs Ri__g and Km vs Delta)
8. Fold proximity diagnostics versus time

3.6 Included Plots

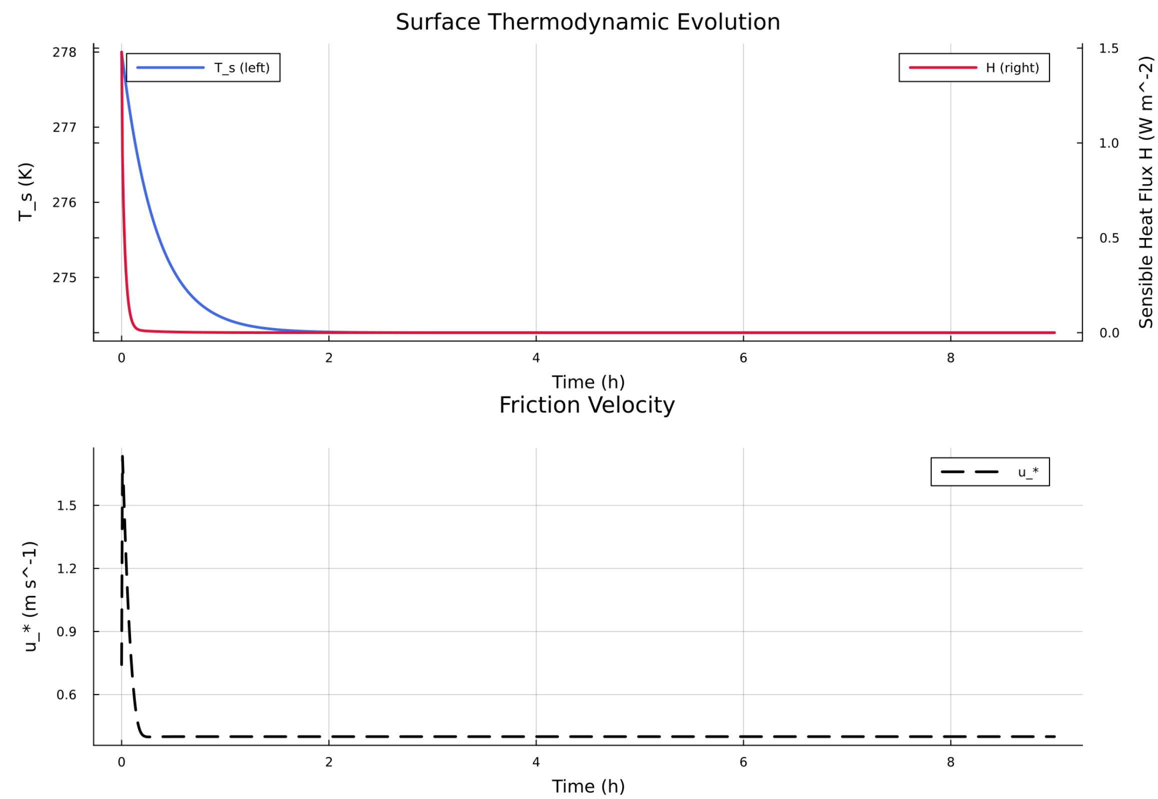

Figure 1: Time series of T__s, sensible heat flux, and friction velocity

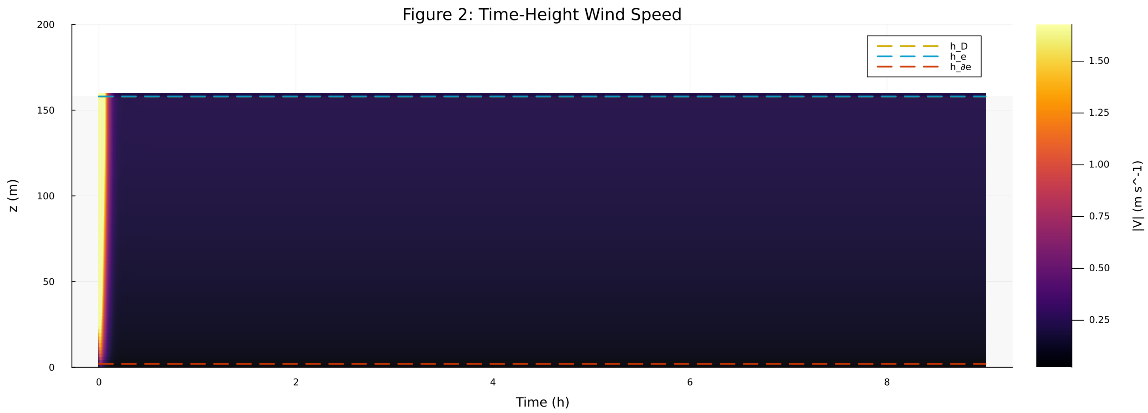

Figure 2: Time-height contour of wind speed with LLJ evolution

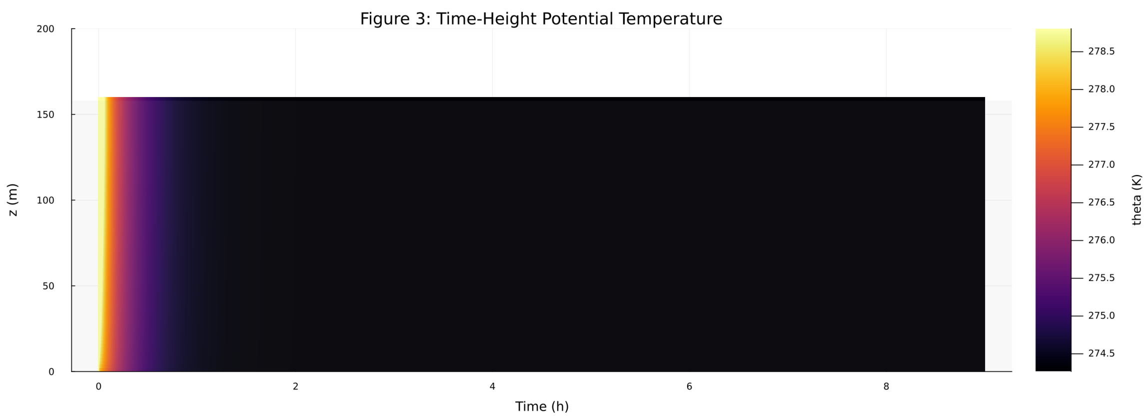

Figure 3: Time-height contour of potential temperature and inversion growth

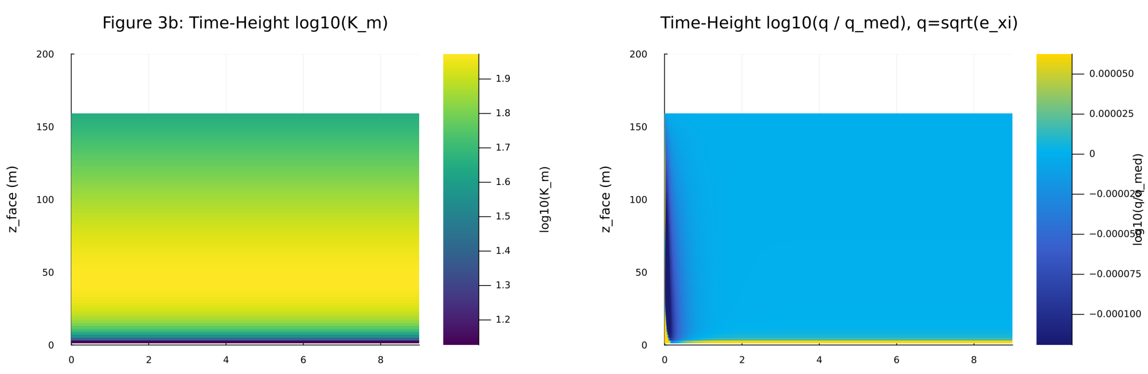

Figure 4: Vertical profiles of U, theta, and Km at representative times

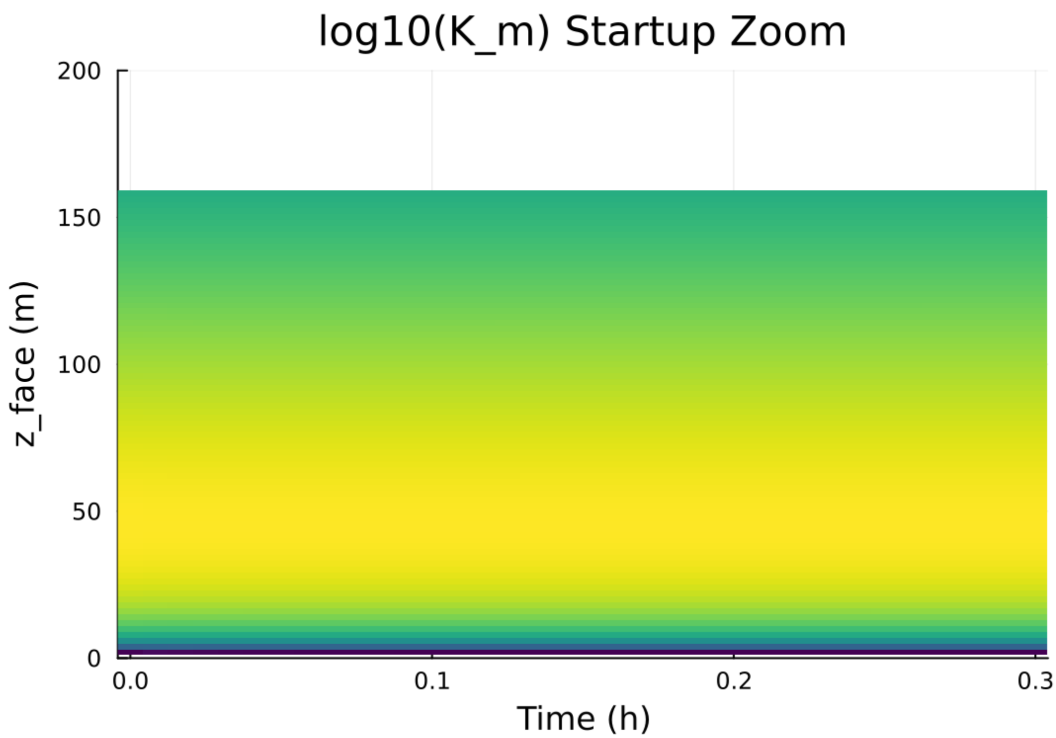

Figure 5: Surface energy budget components (R__n, H, G, storage)

4 Case 2: floss 80x2m 9h report

5 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/floss/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

5.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/floss`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

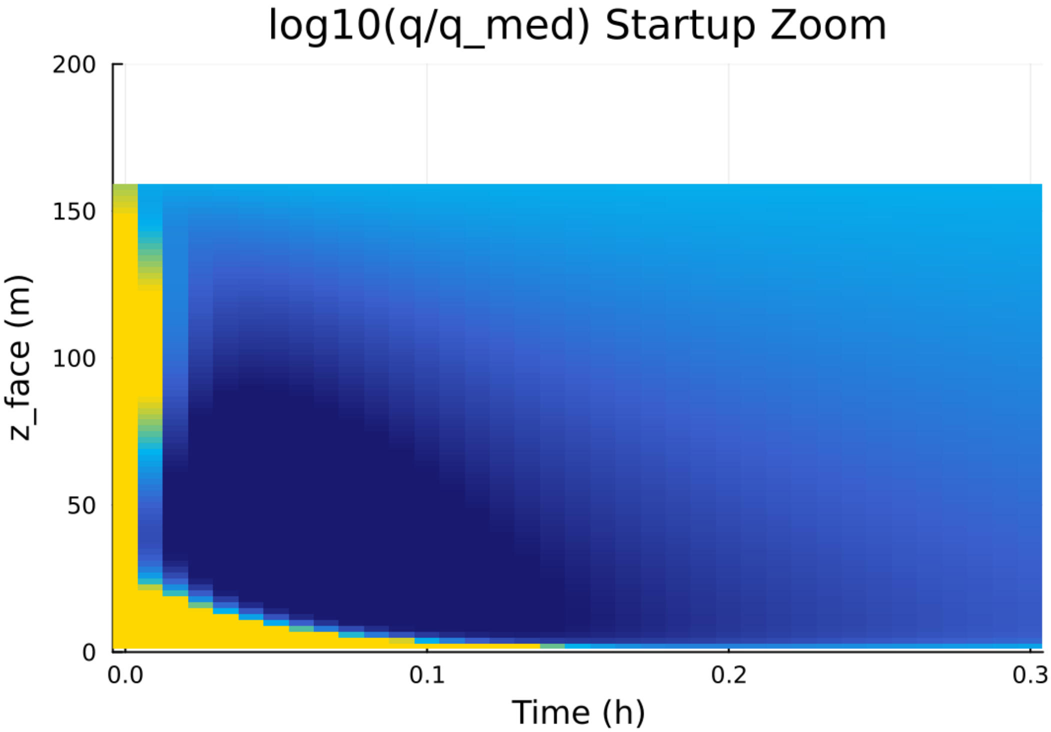

Figure 6: Phase portrait on regularized manifold (Delta vs e_xi)

Table 2: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$10.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	$0.020000\,\mathrm{m}$
Surface roughness (heat)	z_{0h}	$0.005000\,\mathrm{m}$
Coriolis parameter	f	$1.000\mathrm{e}{-04}\,\mathrm{s^{-1}}$
Reference potential temperature	θ_a	$280.000000\,\mathrm{K}$
Deep soil temperature	T_{deep}	$276.000000\,\mathrm{K}$
Downward longwave forcing	$R_\downarrow$	$300.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	$0.100000\,\mathrm{m}$
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	$265.000000\,\mathrm{K}$
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

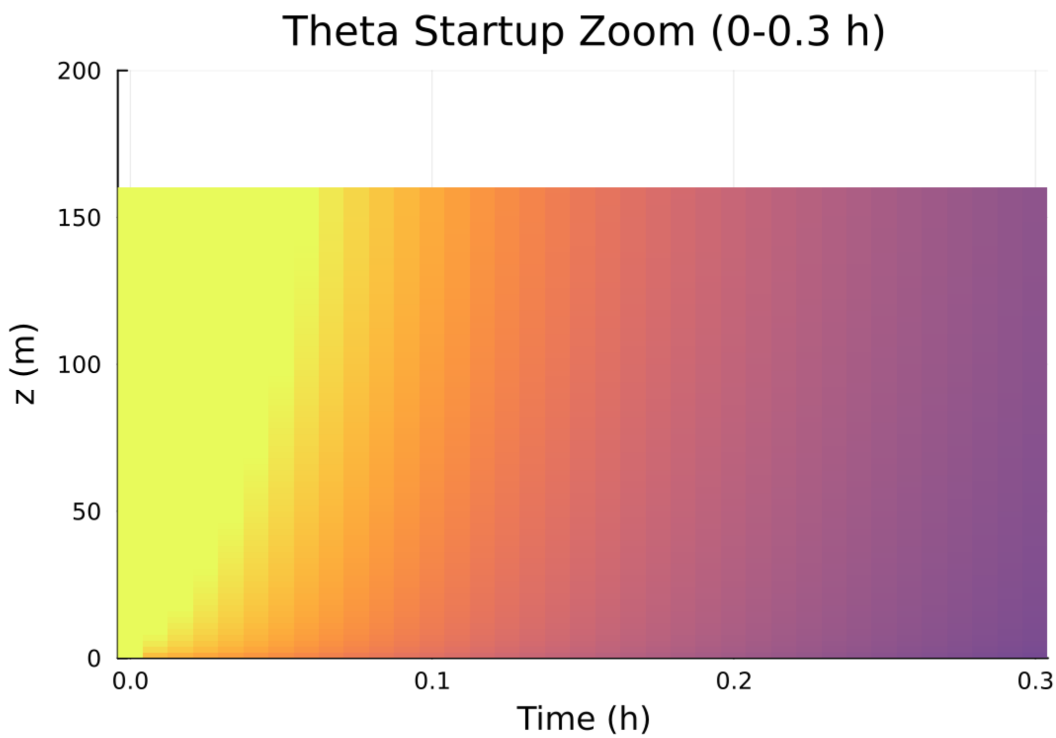

Figure 7: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)

5.2 Forcing and Boundary Conditions

5.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399988, 94.108850]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [-2.012 \times 10^{-6}, 1552.767352]$

5.4 Generated Artifacts

- Time-series CSV: `results/floss/time_series.csv`
- Diagnostic summary JSON: `results/floss/summary.json`
- Payload JLD2: `results/floss/payload.jld2`

5.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

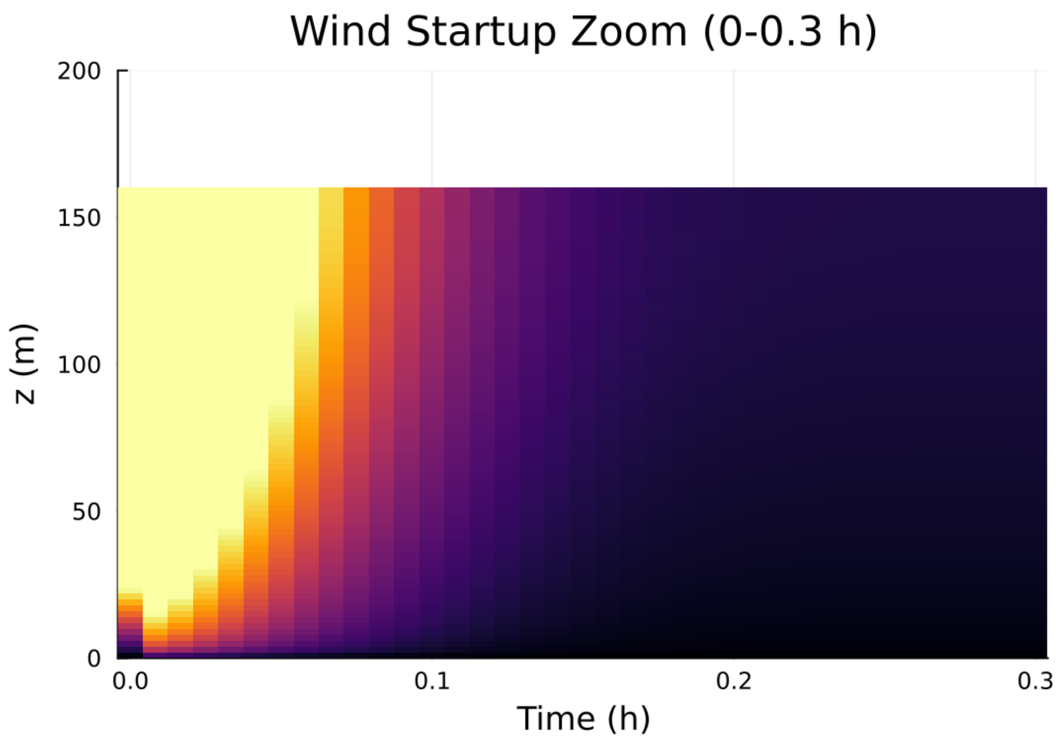

Figure 8: Fold proximity diagnostics versus time

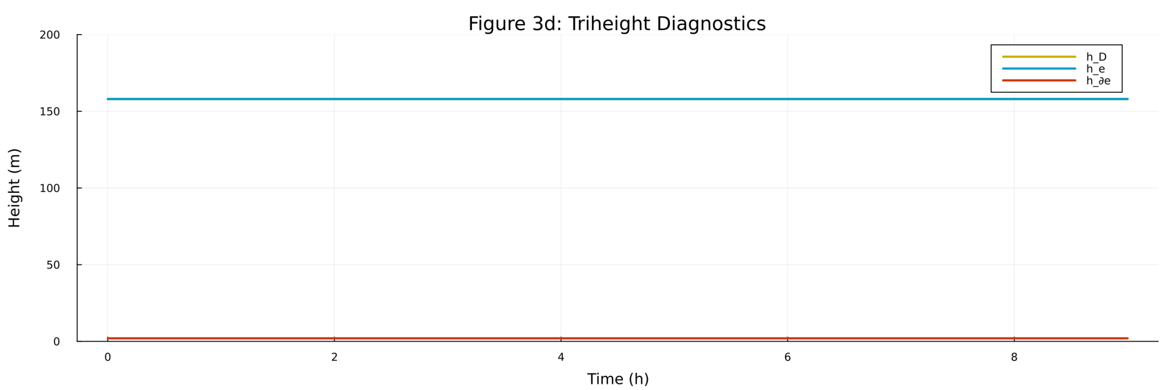

Figure 9: fig03d triheight timeseries.png

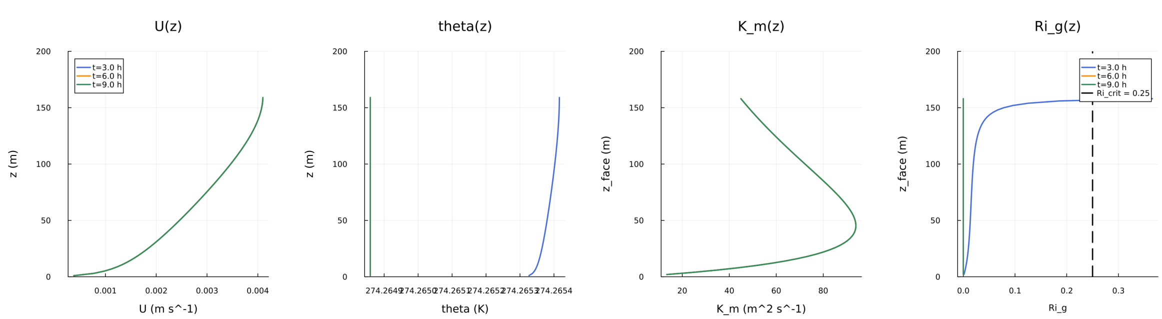

Figure 10: fig04 profiles u theta km.png

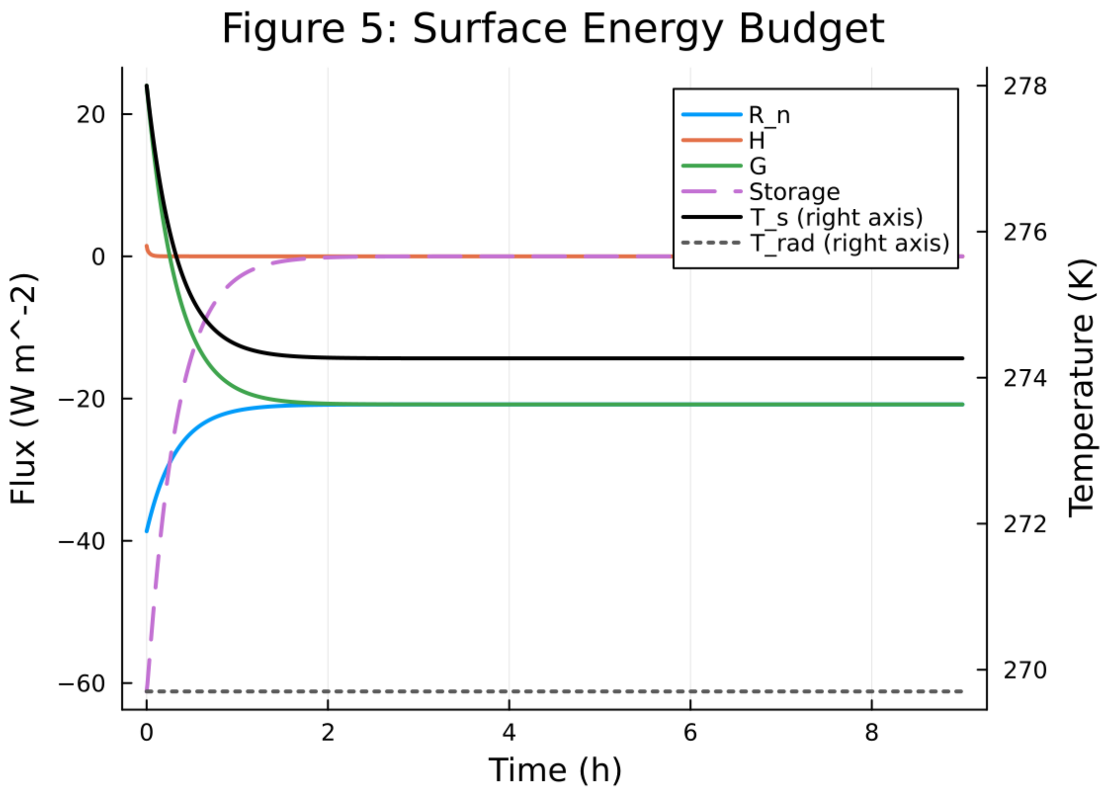

Figure 11: fig05 surface energy budget.png

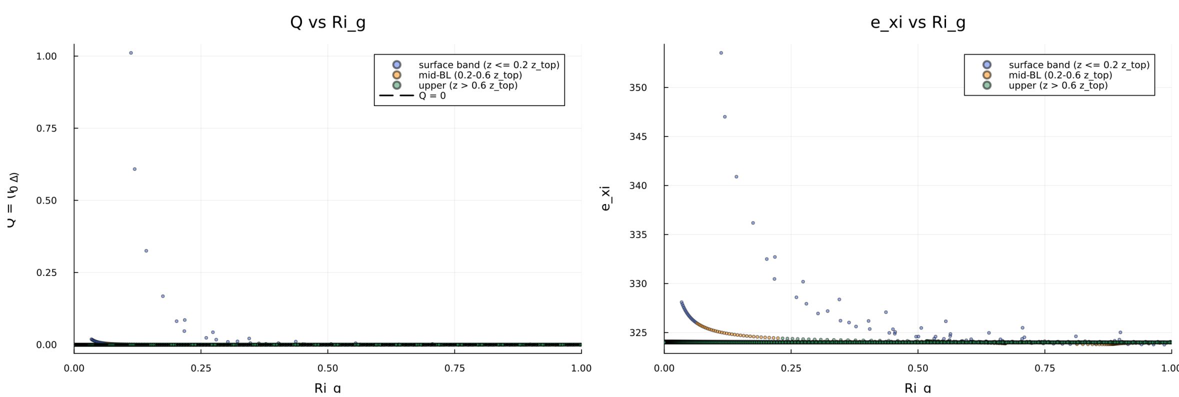

Figure 12: fig06 phase delta exi.png

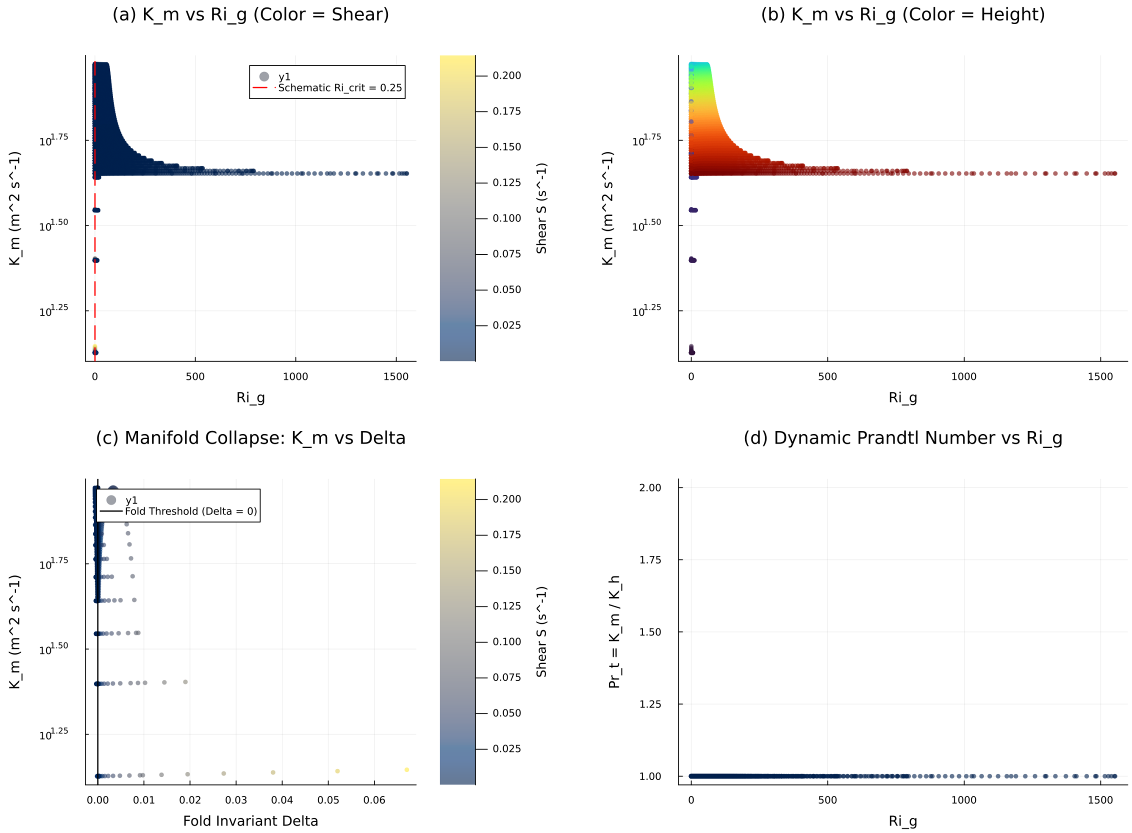

Figure 13: fig07 closure manifold.png

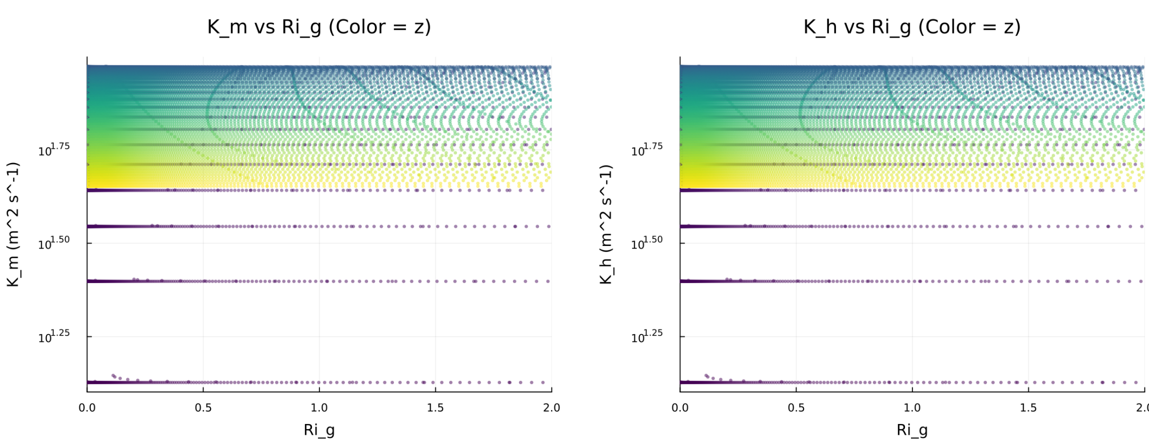

Figure 14: fig07 diffusivity vs ri.png

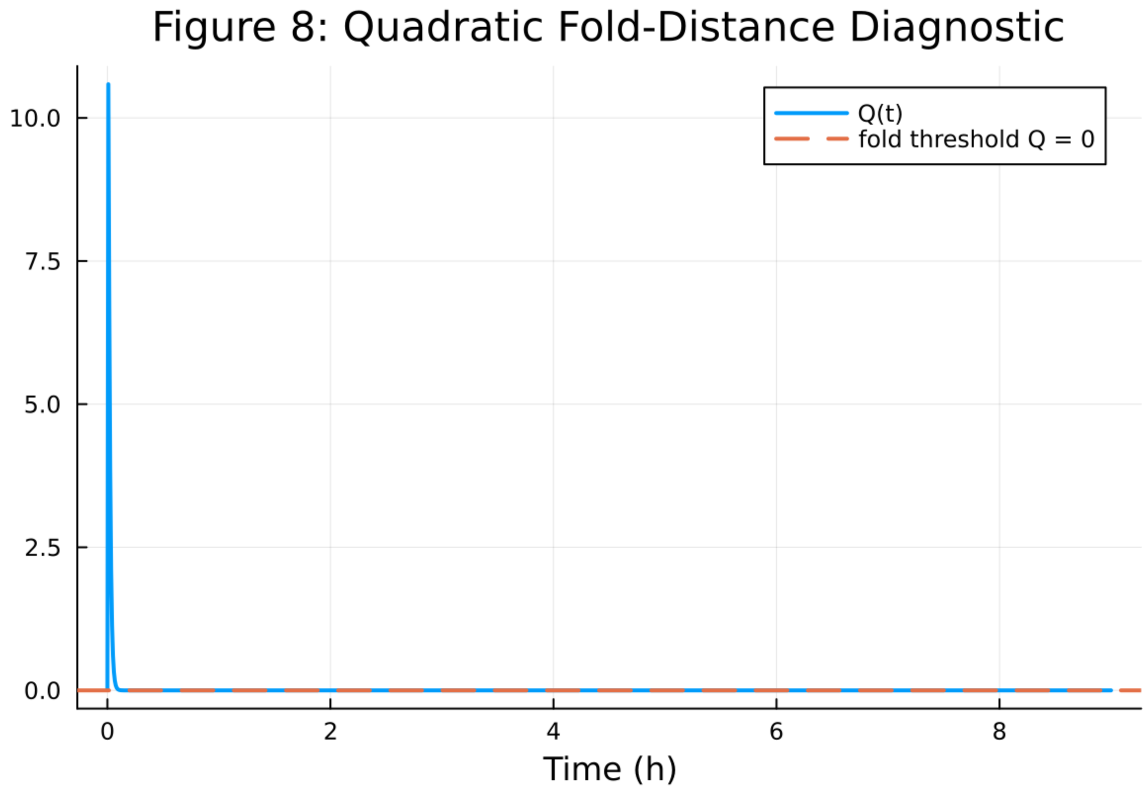

Figure 15: fig08 fold proximity.png

5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Closure manifold diagnostics (Km vs Ri__g and Km vs Delta)
8. Fold proximity diagnostics versus time

5.6 Included Plots

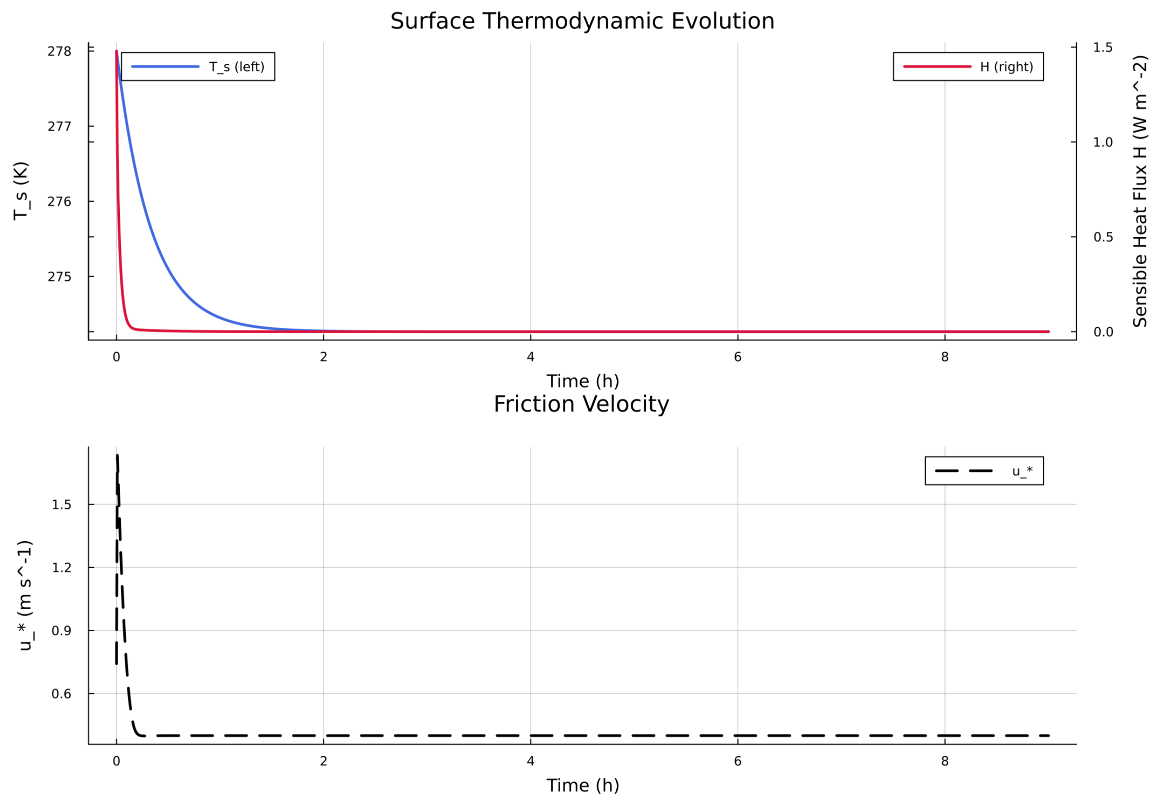

Figure 16: Time series of T__s, sensible heat flux, and friction velocity

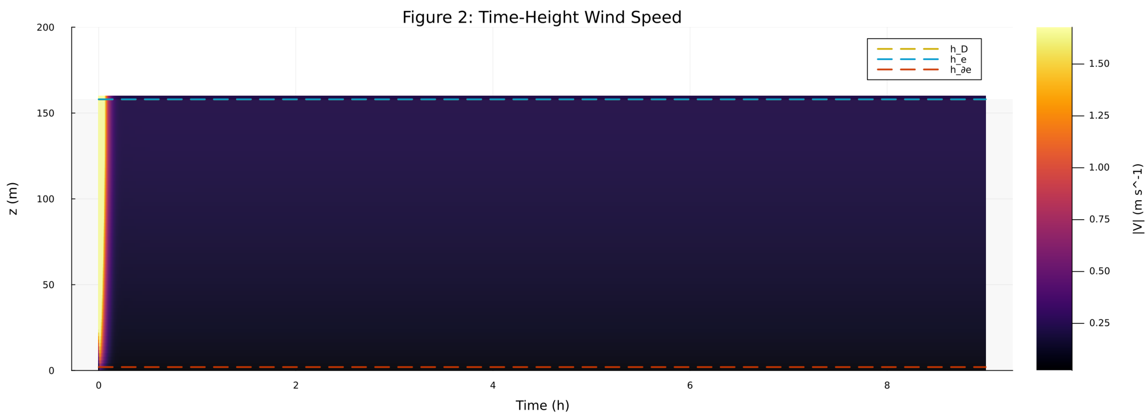

Figure 17: Time-height contour of wind speed with LLJ evolution

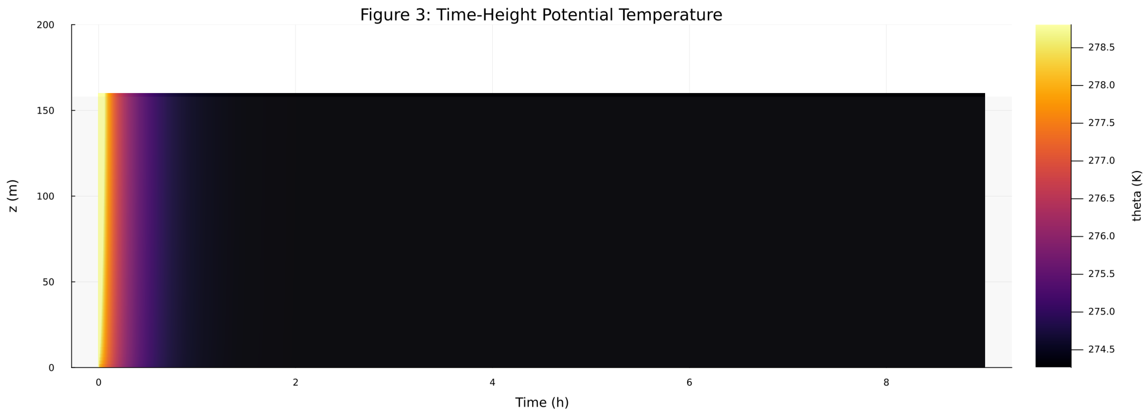

Figure 18: Time-height contour of potential temperature and inversion growth

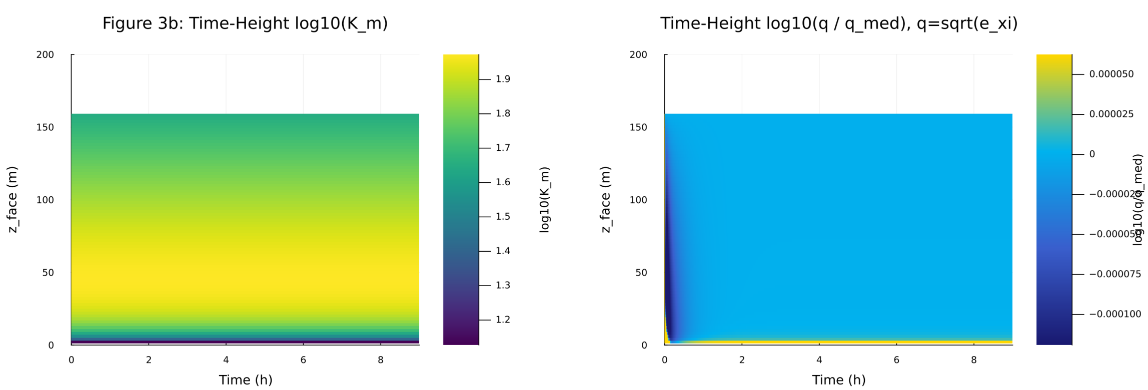

Figure 19: Vertical profiles of U, theta, and Km at representative times

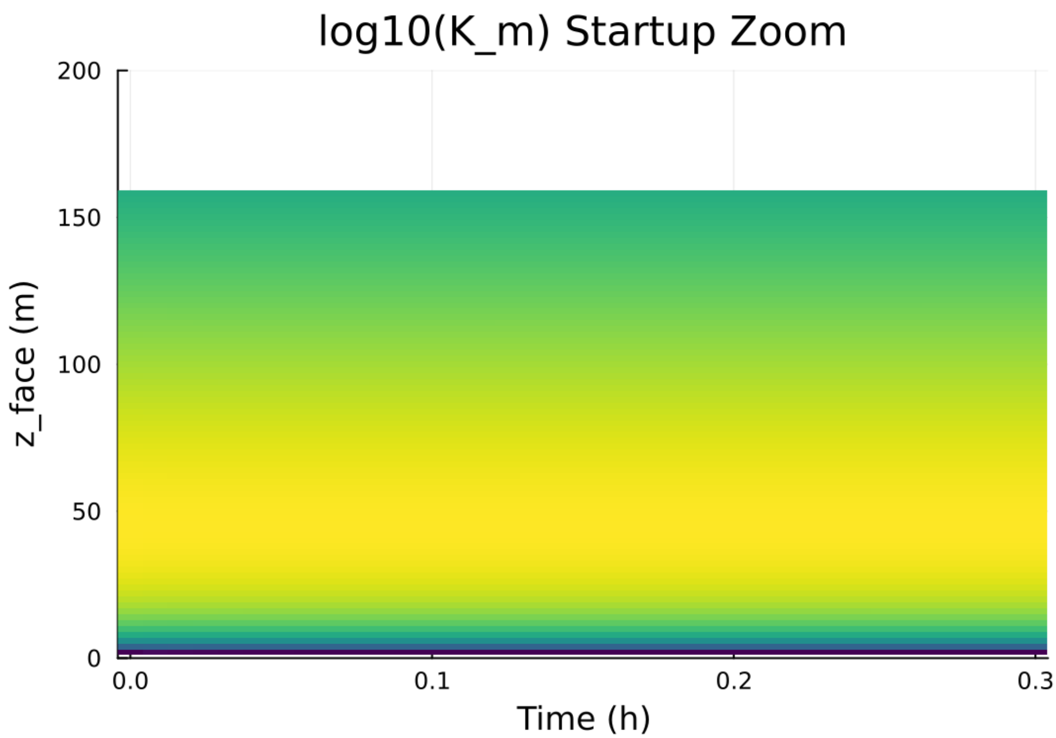

Figure 20: Surface energy budget components (R_n, H, G, storage)

6 Case 3: sheba 160x2m 12h report

7 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

7.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 100 / 0
- RHS evaluations: 49002
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

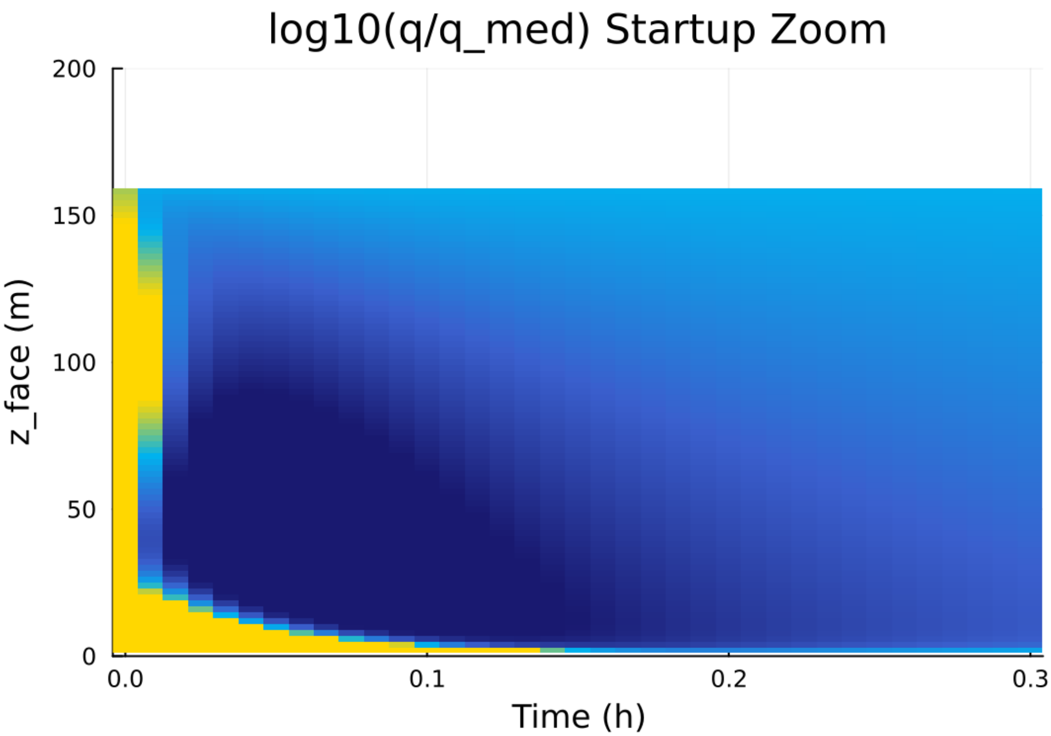

Figure 21: Phase portrait on regularized manifold (Delta vs e_xi)

Table 3: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	7.000000 m s^{-1}
Meridional geostrophic wind	V_g	0.000000 m s^{-1}
Surface roughness (momentum)	z_{0m}	1.000e-04 m
Surface roughness (heat)	z_{0h}	1.000e-05 m
Coriolis parameter	f	1.000e-04 s^{-1}
Reference potential temperature	θ_a	257.000000 K
Deep soil temperature	T_{deep}	255.500000 K
Downward longwave forcing	$R_\downarrow$	190.000000 W m^{-2}
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\min,\mathrm{surf}}$	0.001000 m^2 s^{-1}
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	257.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

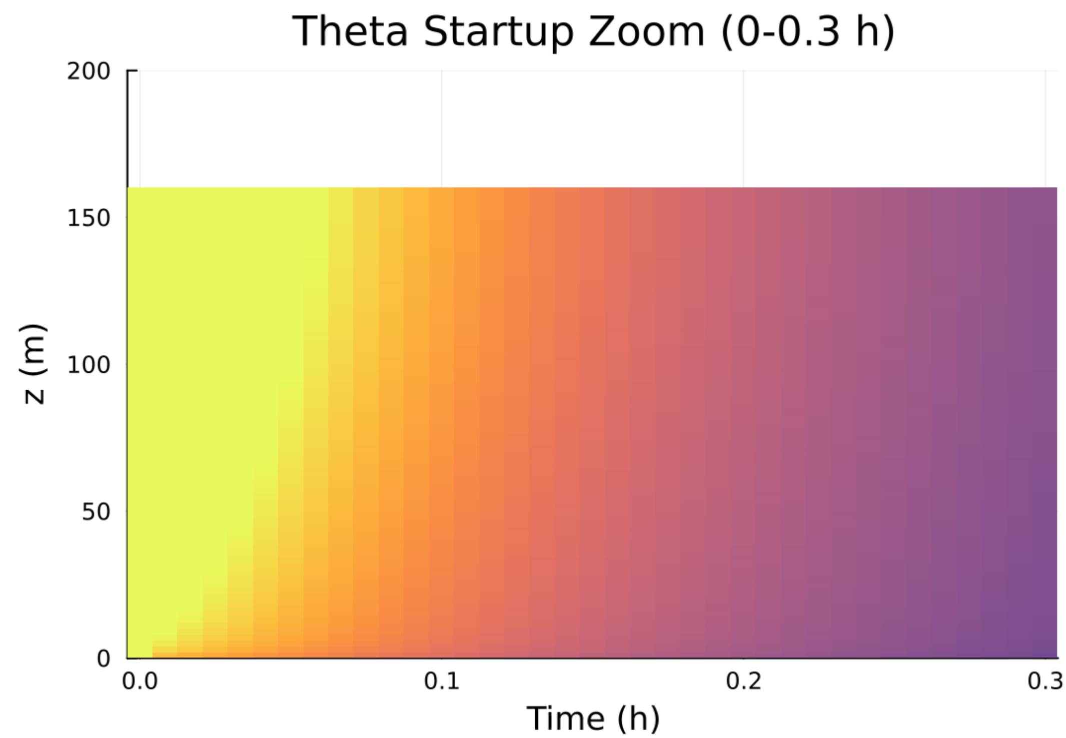

Figure 22: Closure manifold diagnostics (Km vs Ri_g and Km vs Delta)

7.2 Forcing and Boundary Conditions

7.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [13.028339, 141.433744]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [-7.243 \times 10^{-6}, 3569.000203]$

7.4 Generated Artifacts

- Time-series CSV: `results/sheba/time_series.csv`
- Diagnostic summary JSON: `results/sheba/summary.json`
- Payload JLD2: `results/sheba/payload.jld2`

7.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

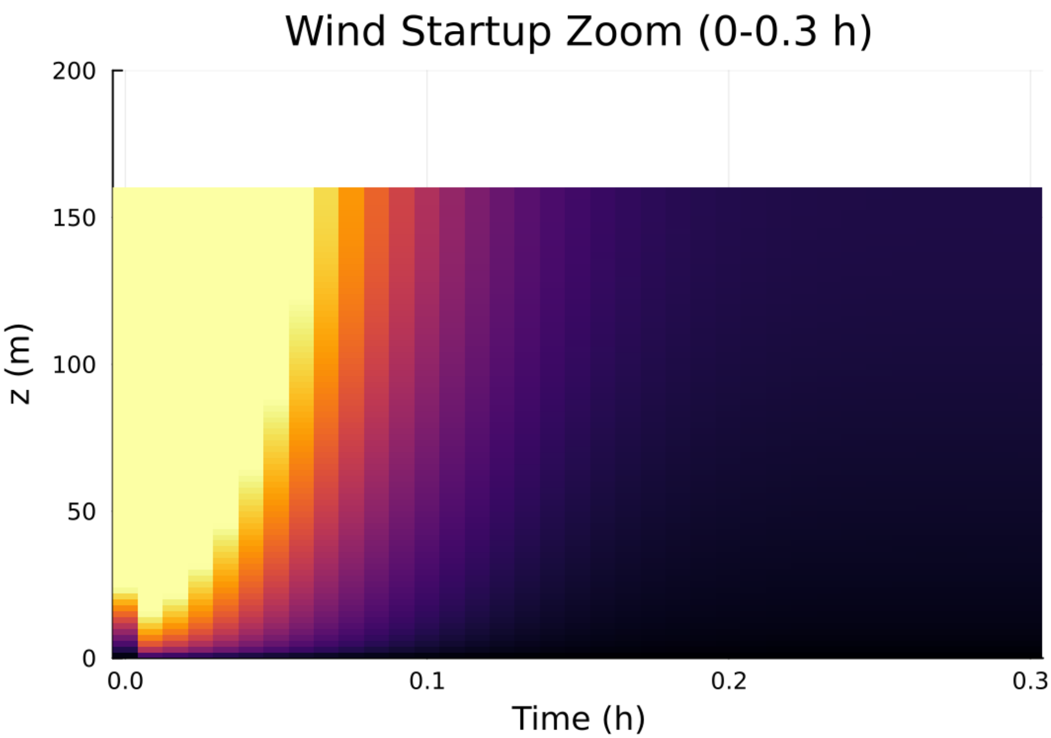

Figure 23: Fold proximity diagnostics versus time

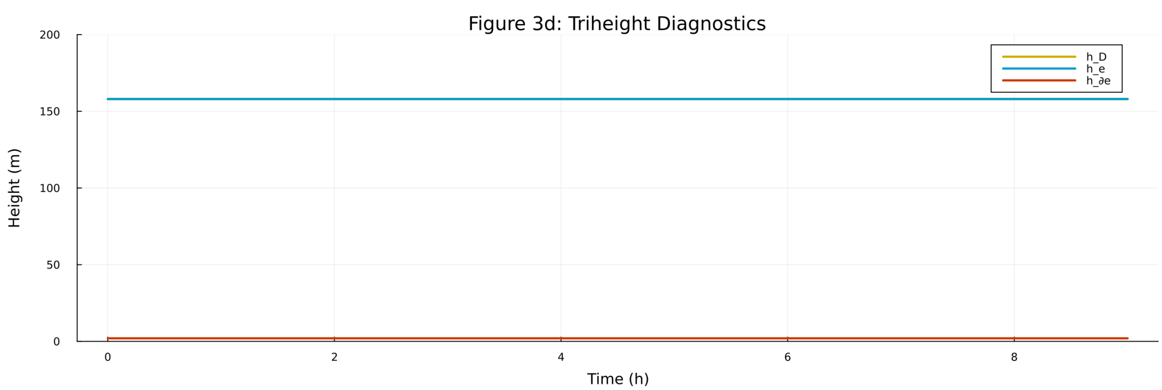

Figure 24: fig03d triheight timeseries.png

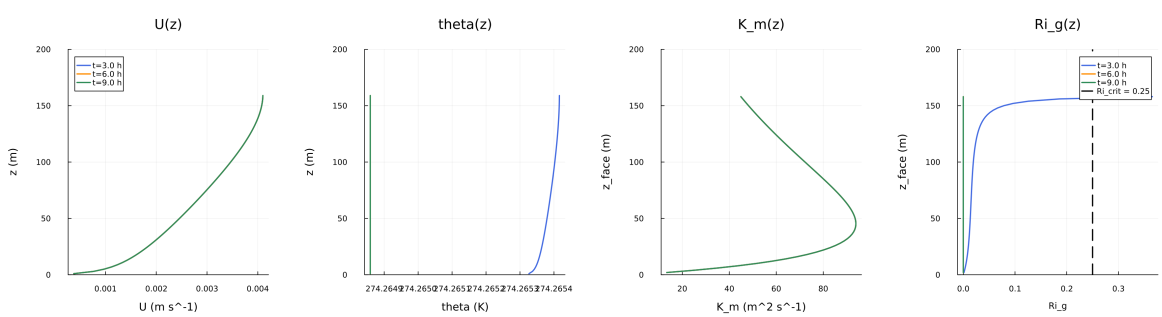

Figure 25: fig04 profiles u theta km.png

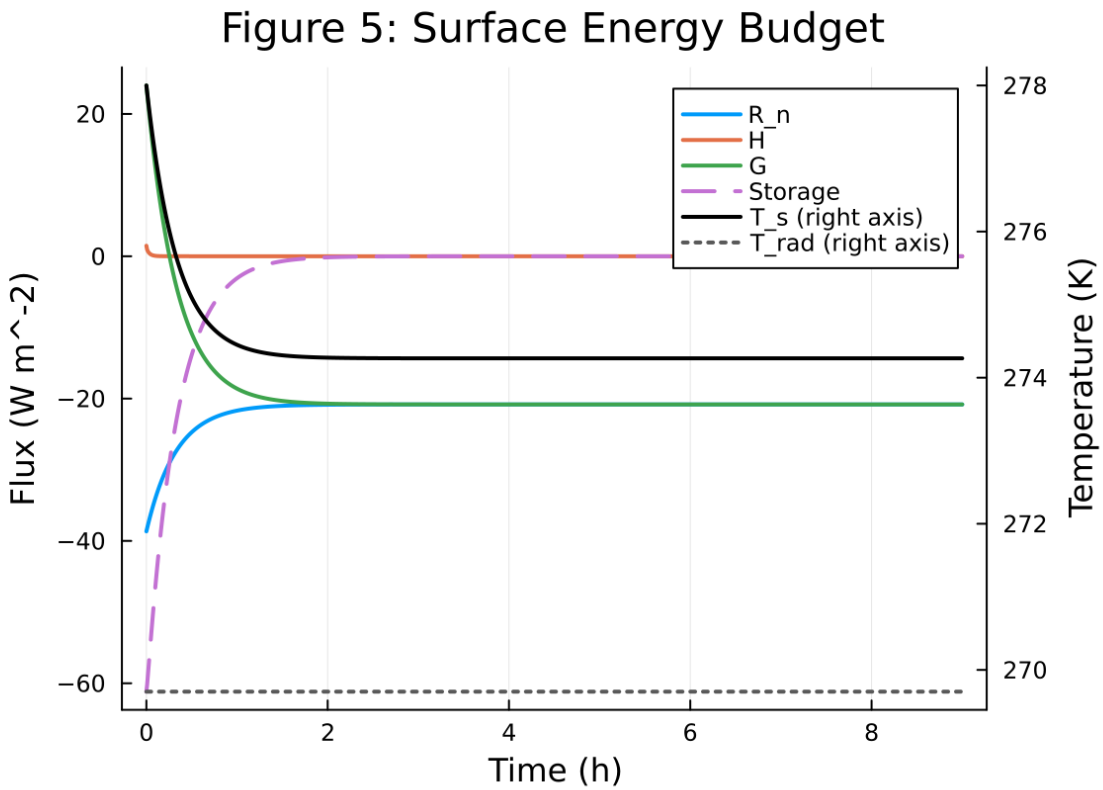

Figure 26: fig05 surface energy budget.png

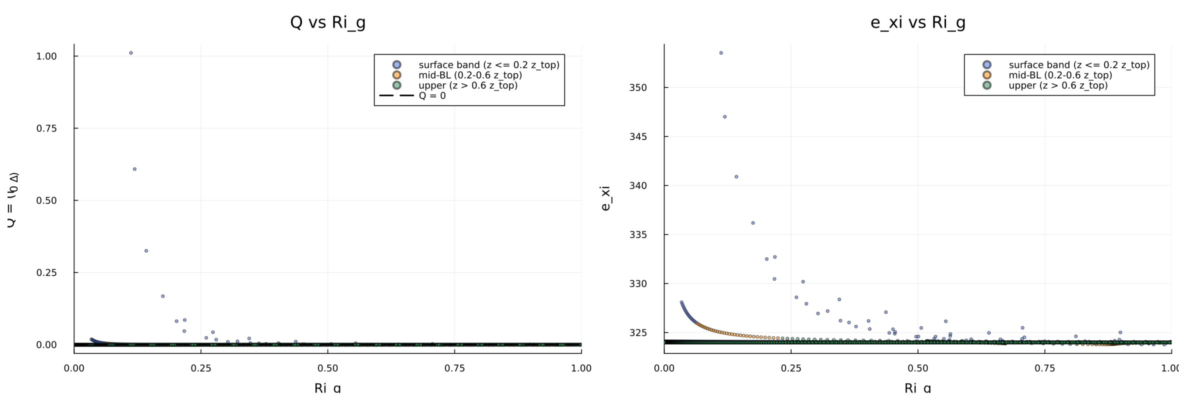

Figure 27: fig06 phase delta exi.png

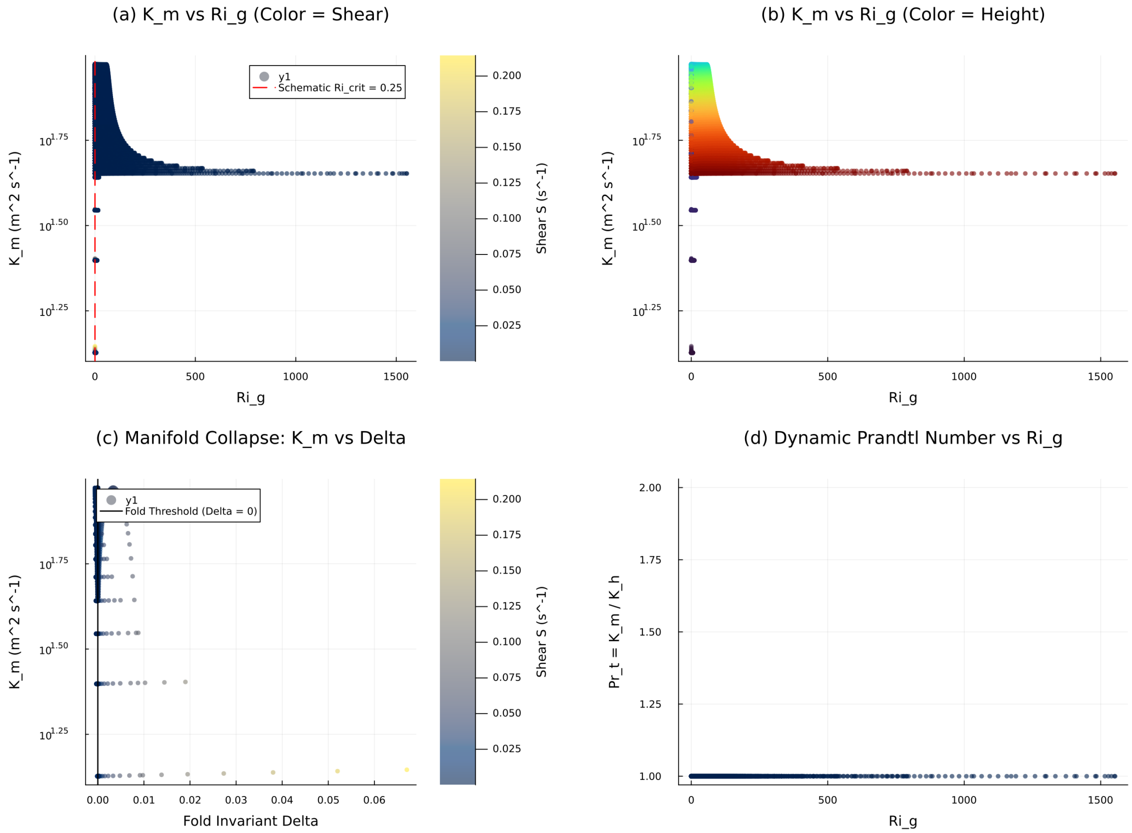

Figure 28: fig07 closure manifold.png

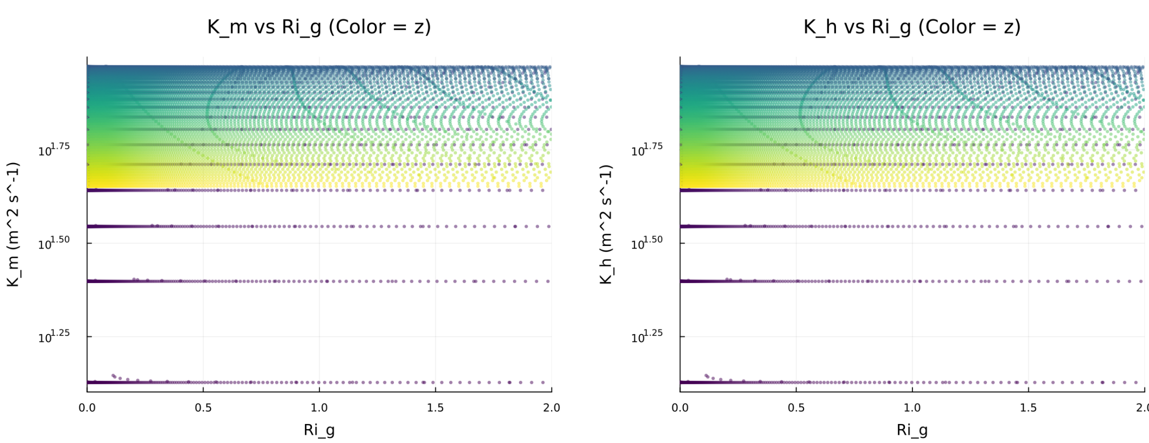

Figure 29: fig07 diffusivity vs ri.png

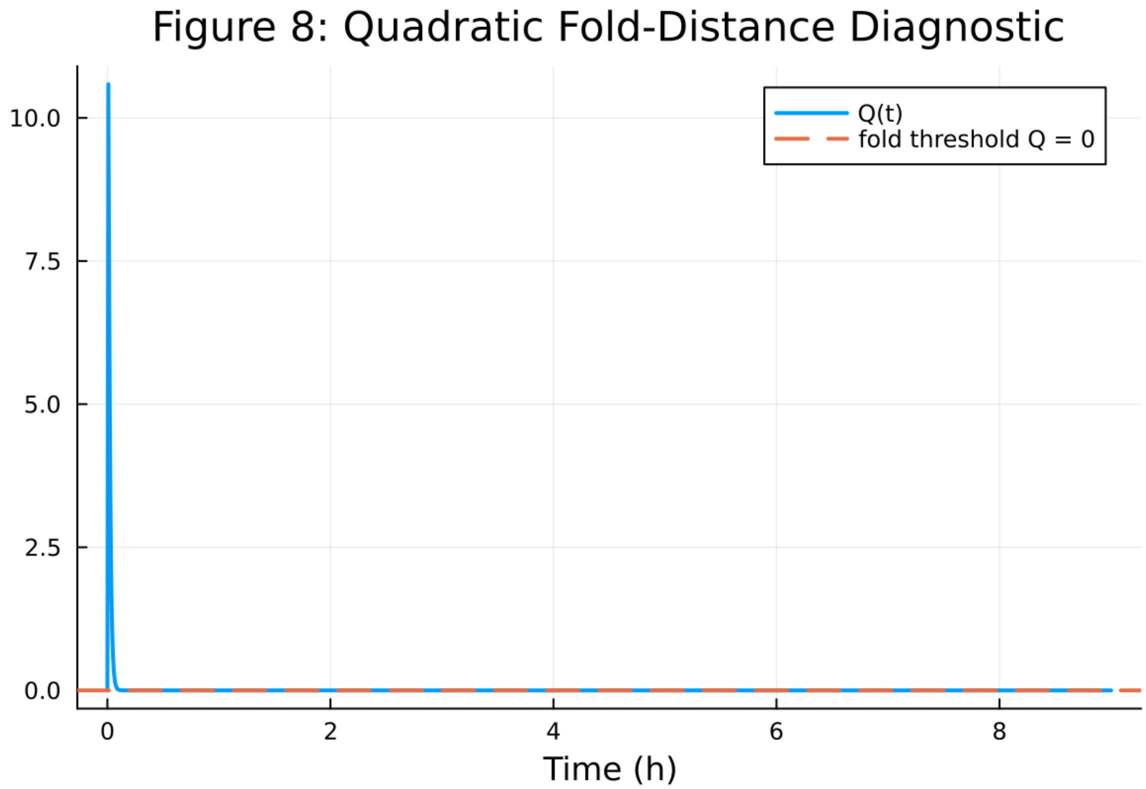

Figure 30: fig08 fold proximity.png

5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

7.6 Included Plots

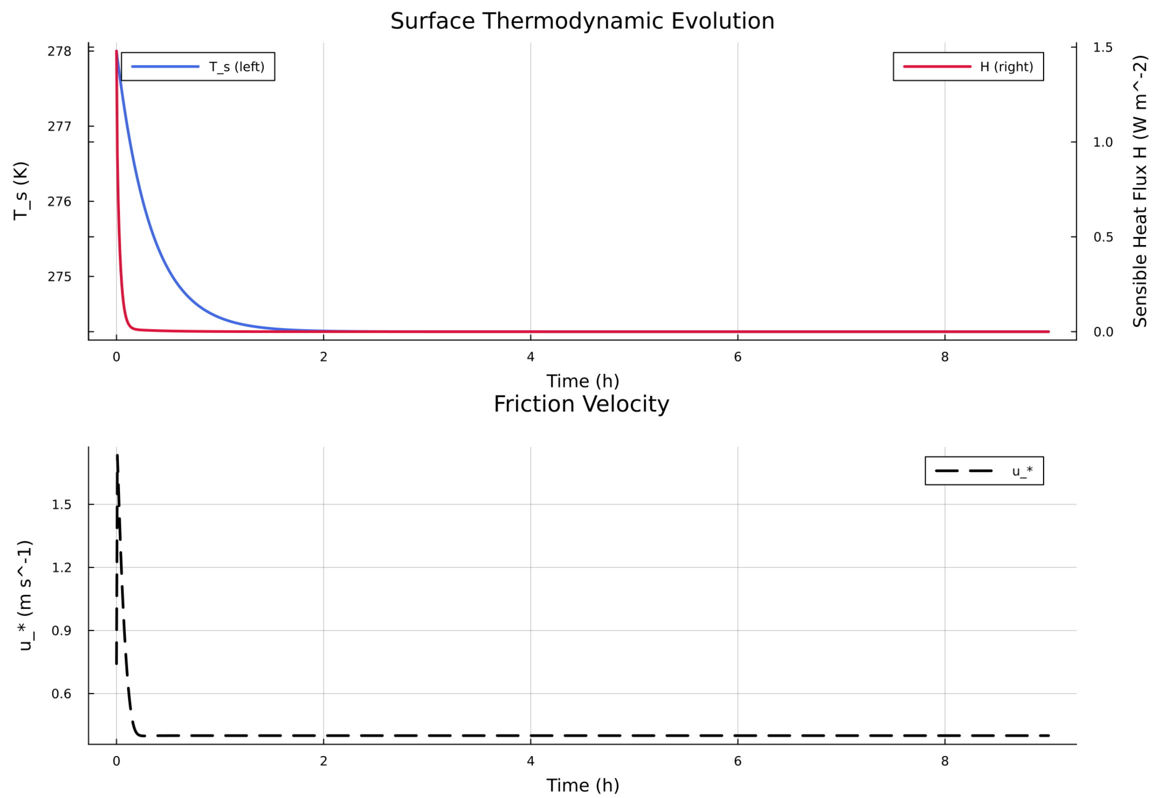

Figure 31: Time series of T__s, sensible heat flux, and friction velocity

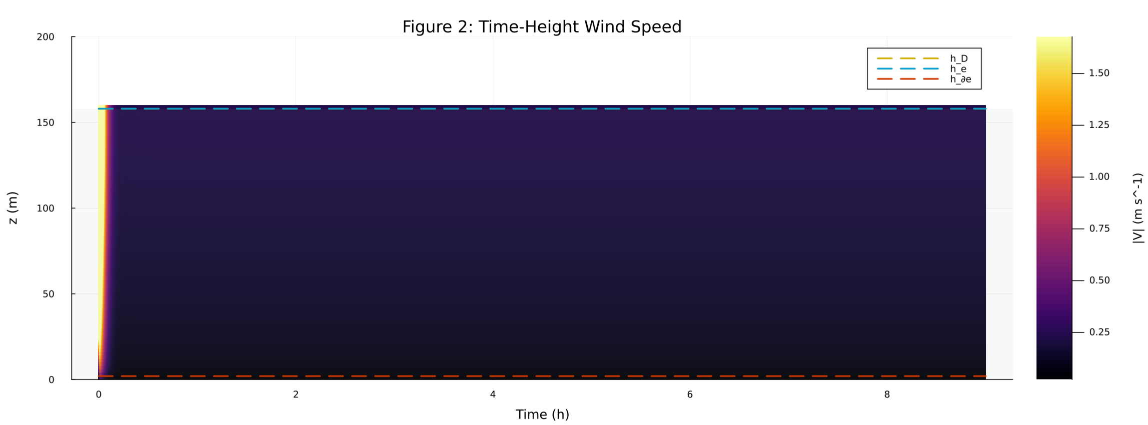

Figure 32: Time-height contour of wind speed with LLJ evolution

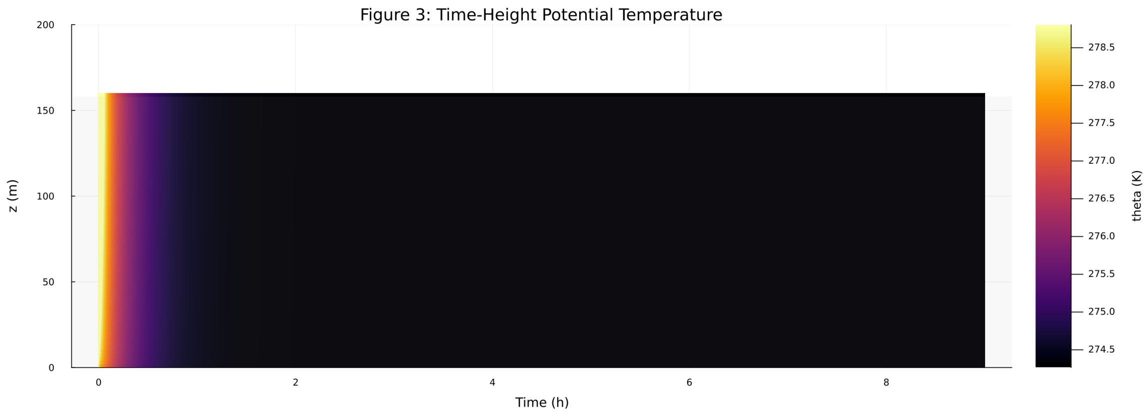

Figure 33: Time-height contour of potential temperature and inversion growth

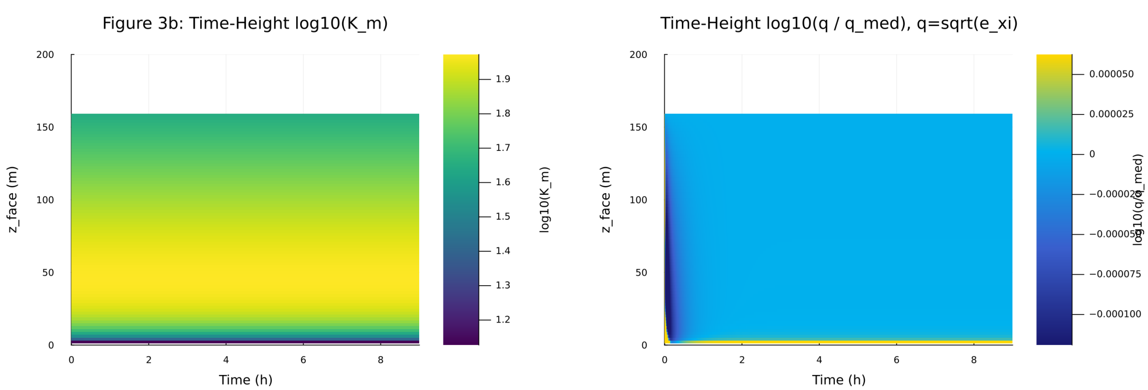

Figure 34: Vertical profiles of U, theta, and Km at representative times

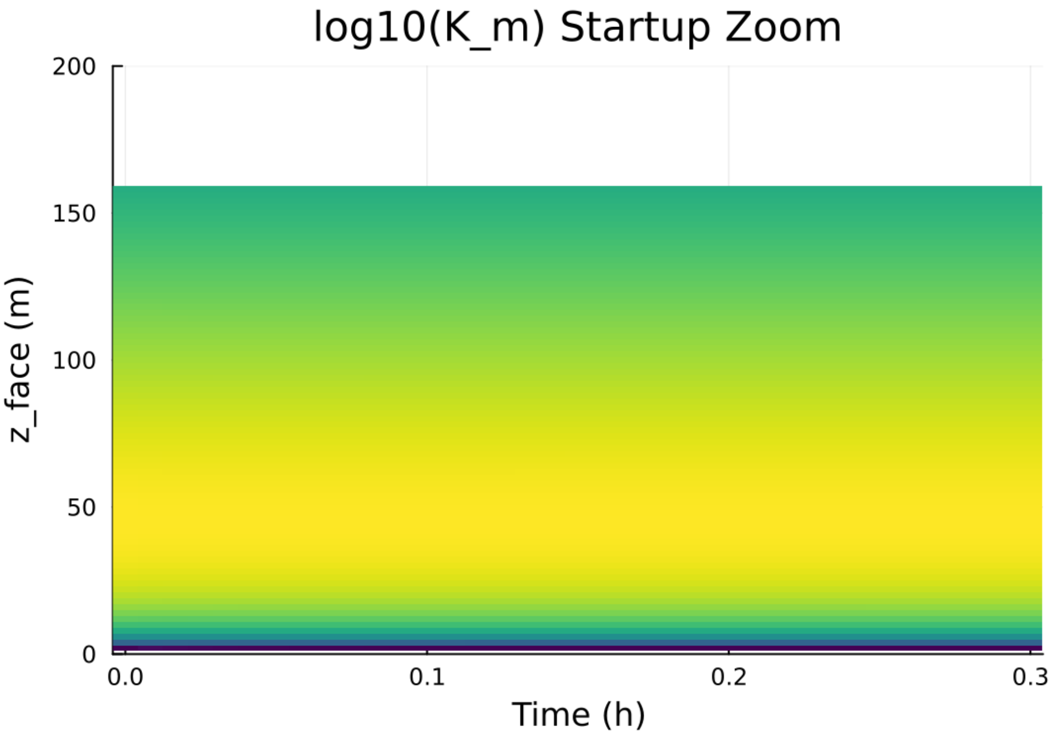

Figure 35: Surface energy budget components (R_n, H, G, storage)

8 Case 4: gabls1 200x2m 12h report

9 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

9.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1441
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 99 / 0
- RHS evaluations: 60392
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

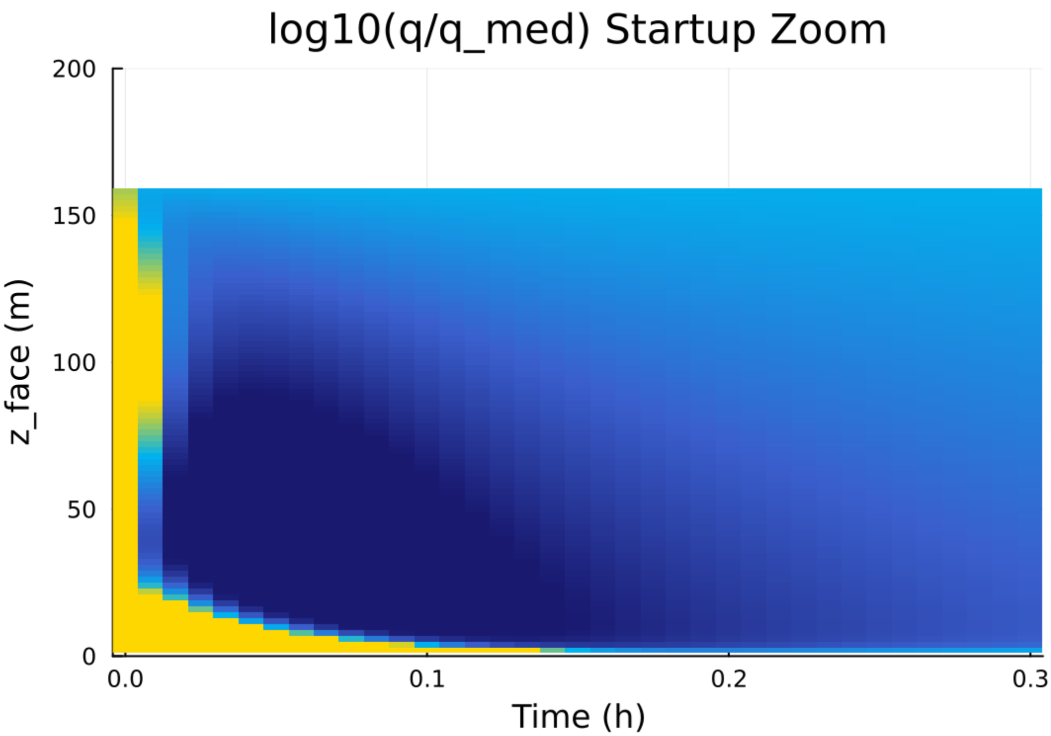

Figure 36: Phase portrait on regularized manifold (Delta vs e_xi)

Table 4: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	8.000000 $\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	0.000000 $\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	0.100000 m
Surface roughness (heat)	z_{0h}	0.010000 m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	245.000000 $\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 $\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 $\mathrm{s^{-1}}$

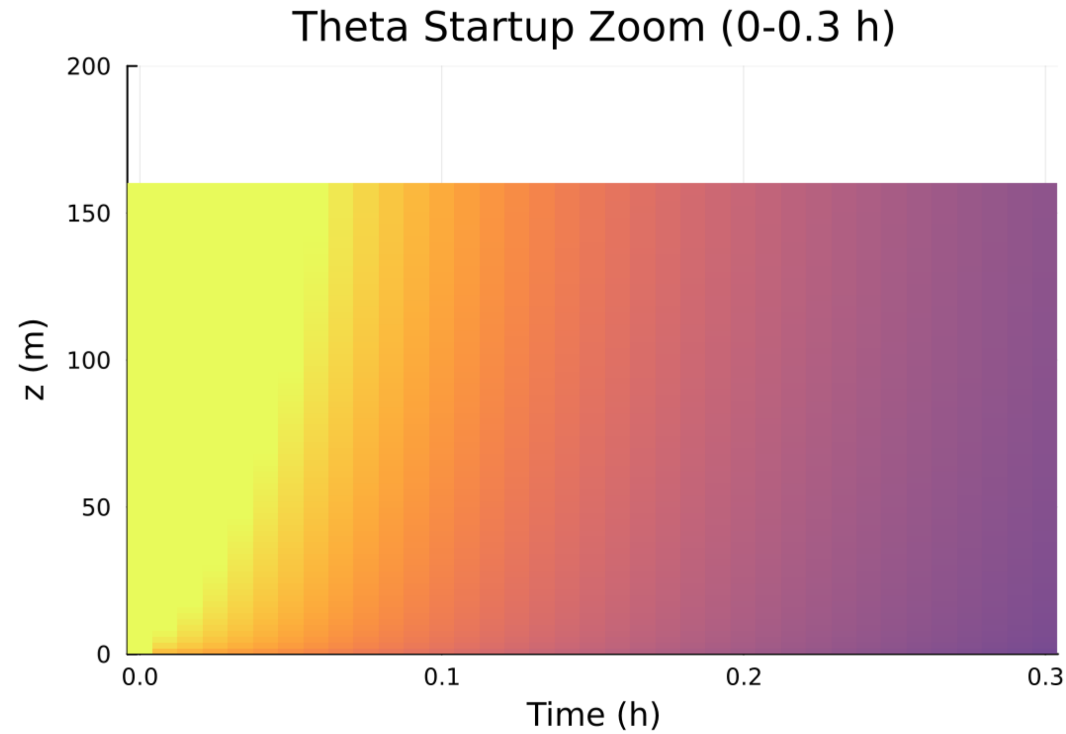

Figure 37: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

9.2 Forcing and Boundary Conditions

9.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [3.844877, 78.288180]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [6.468 \times 10^{-9}, 843.105654]$

9.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

9.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

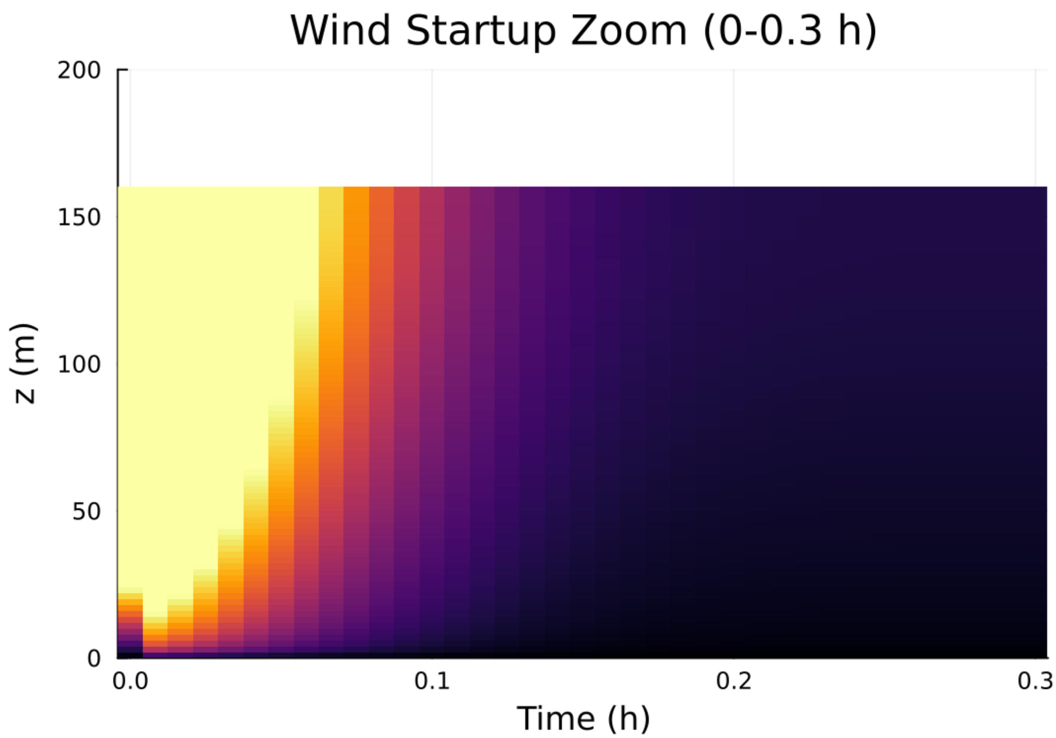

Figure 38: Fold proximity diagnostics versus time

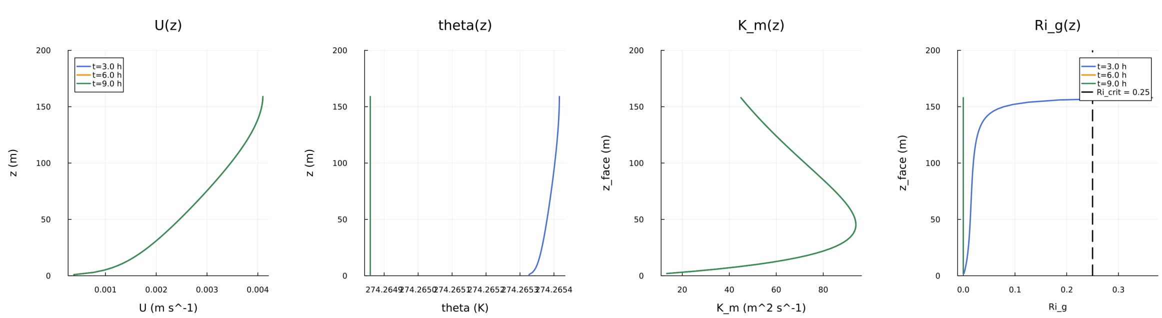

Figure 39: fig04 profiles u theta km.png

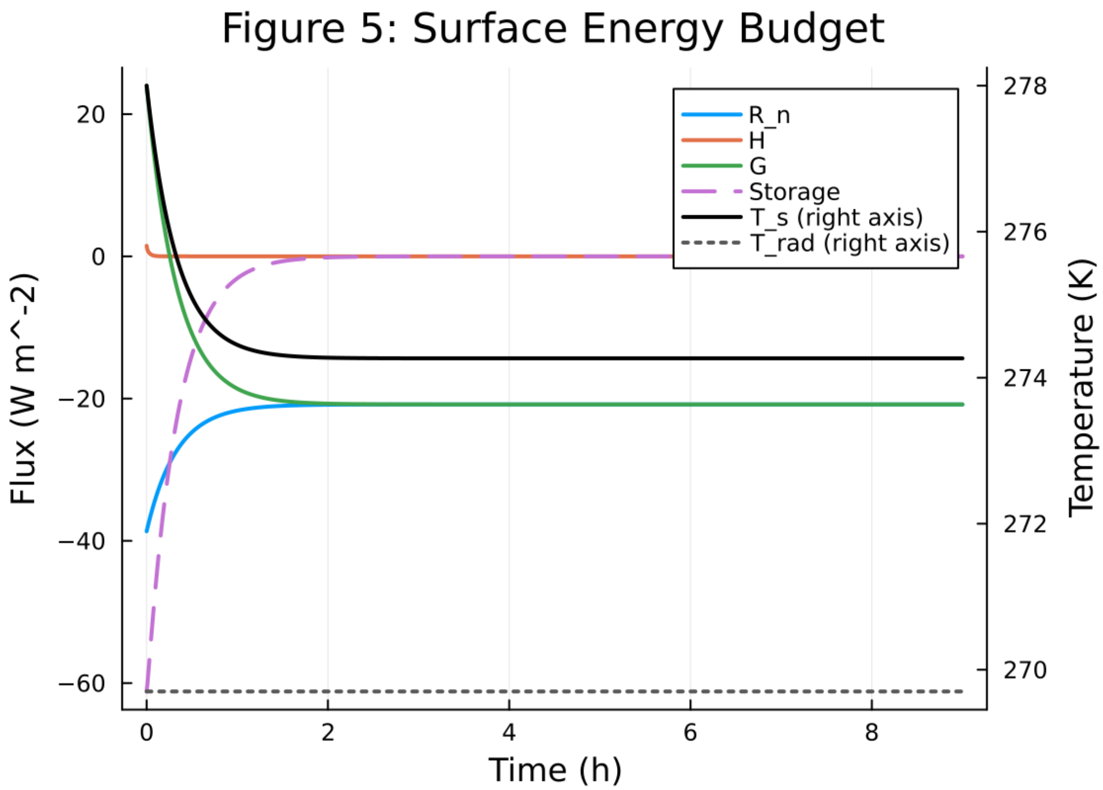

Figure 40: fig05 surface energy budget.png

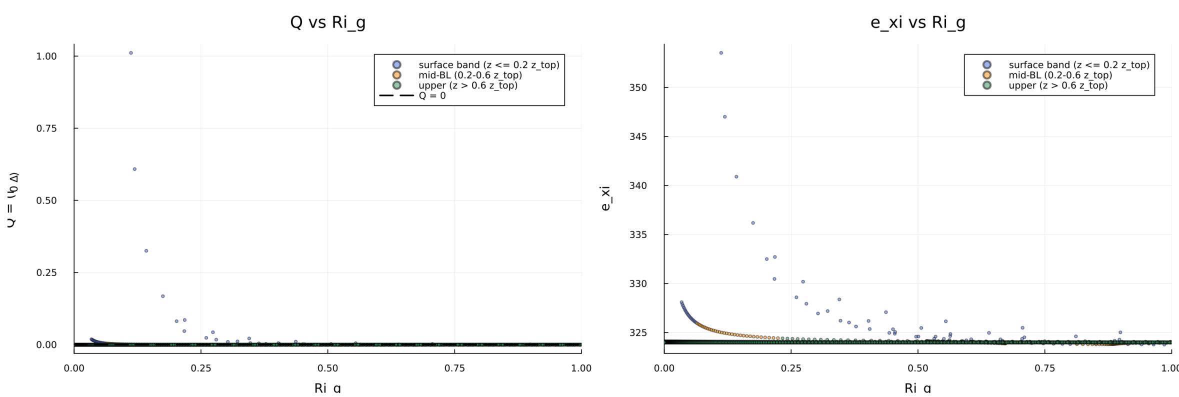

Figure 41: fig06 phase delta exi.png

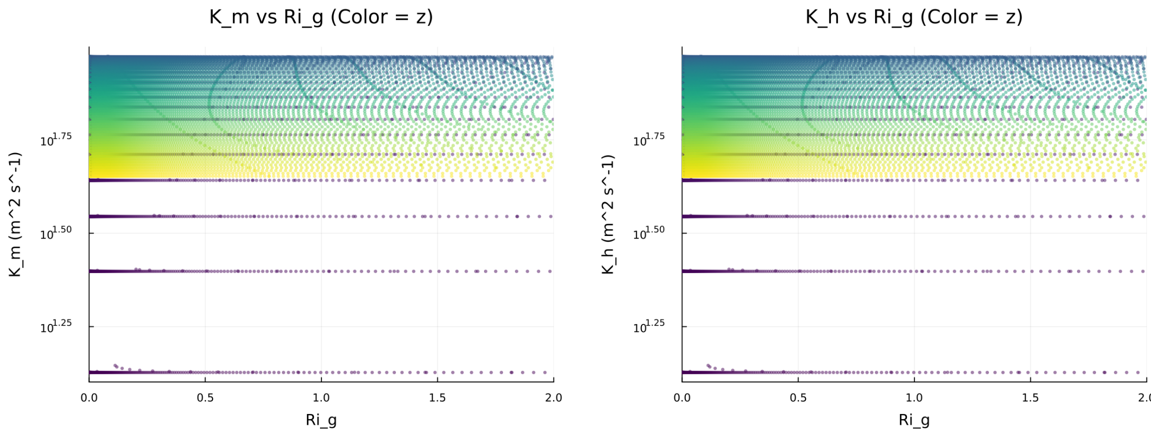

Figure 42: fig07 diffusivity vs ri.png

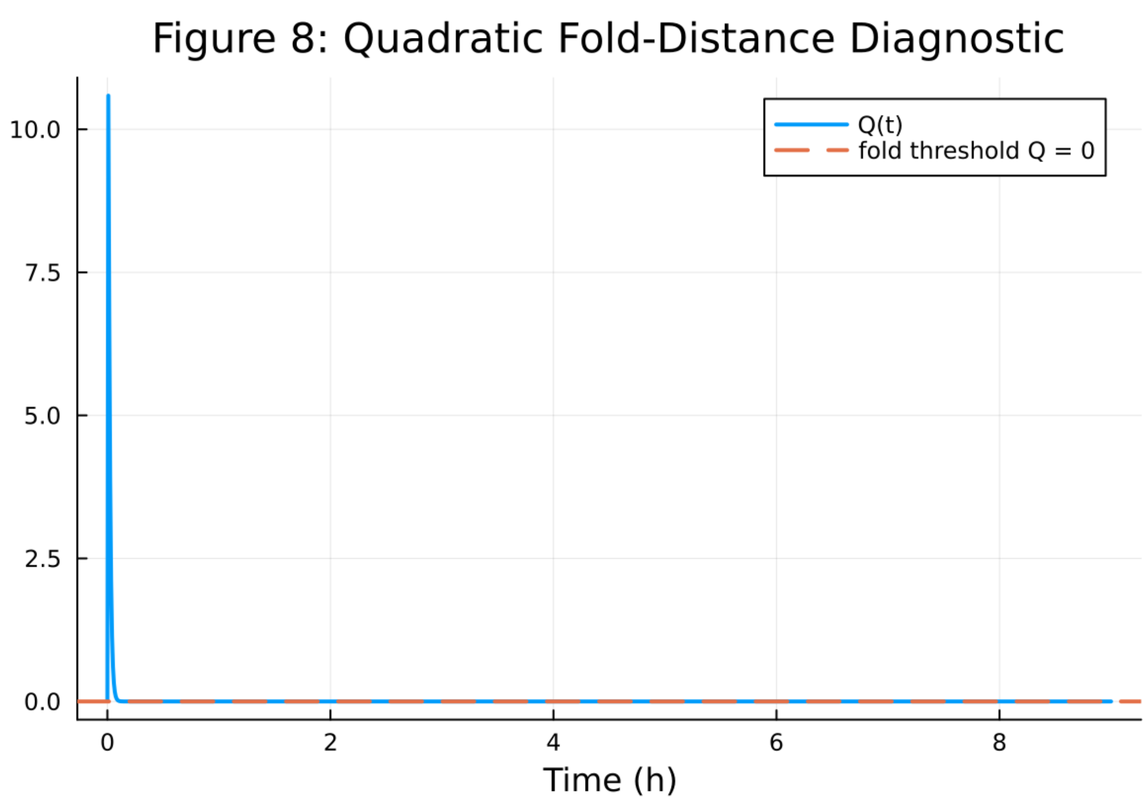

Figure 43: fig08 fold proximity.png

5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

9.6 Included Plots

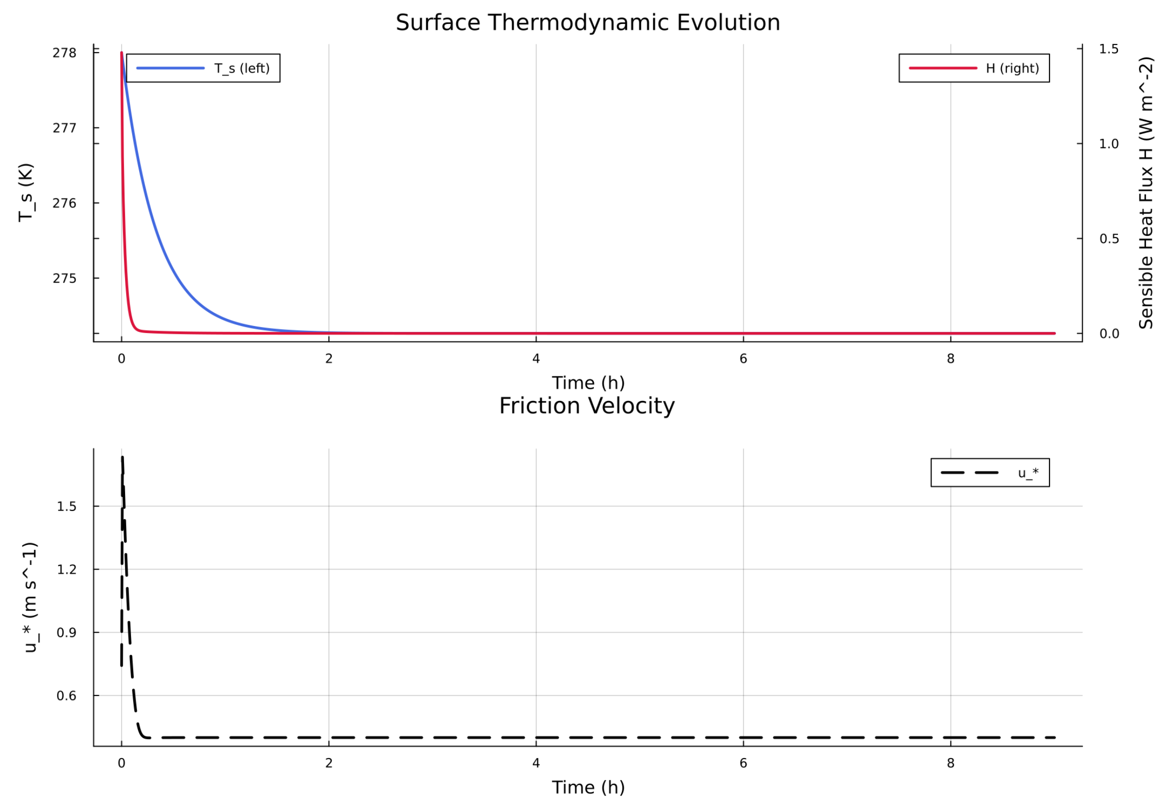

Figure 44: Time series of T__s, sensible heat flux, and friction velocity

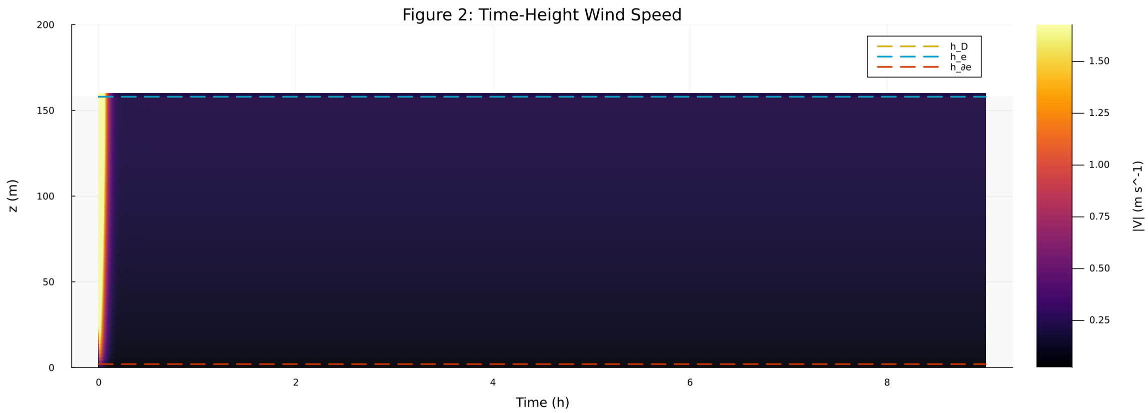

Figure 45: Time-height contour of wind speed with LLJ evolution

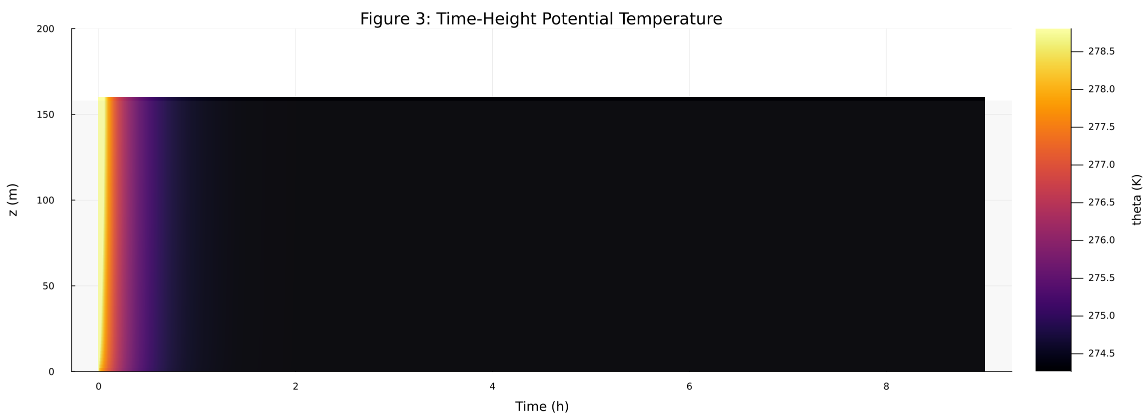

Figure 46: Time-height contour of potential temperature and inversion growth

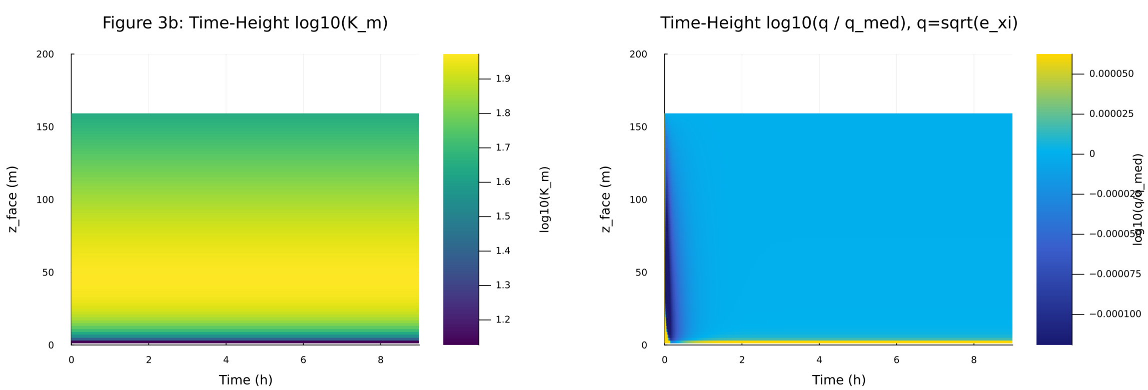

Figure 47: Vertical profiles of U, theta, and Km at representative times

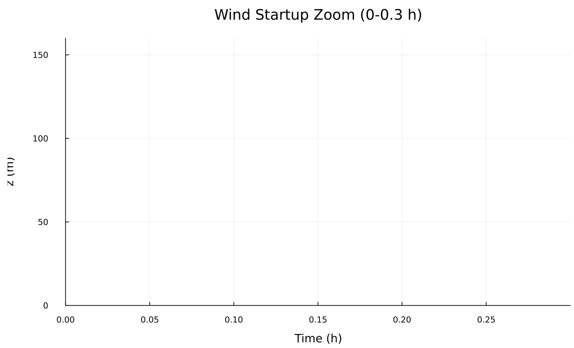

Figure 48: Surface energy budget components (R__n, H, G, storage)

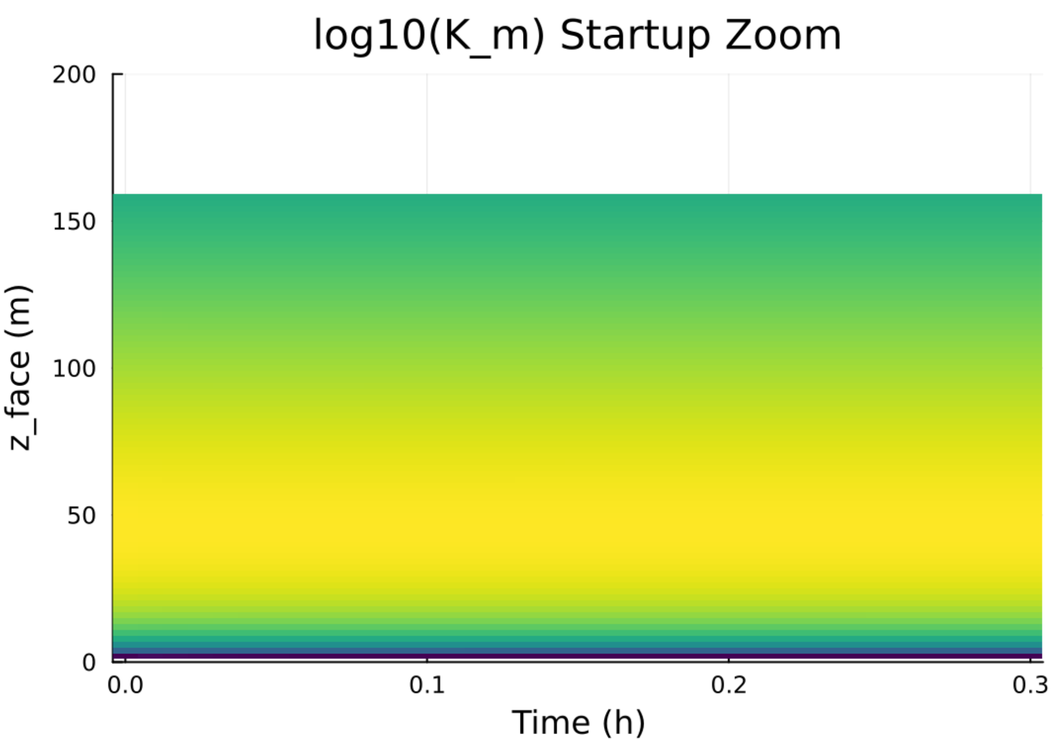

Figure 49: Phase portrait on regularized manifold (Delta vs e__xi)

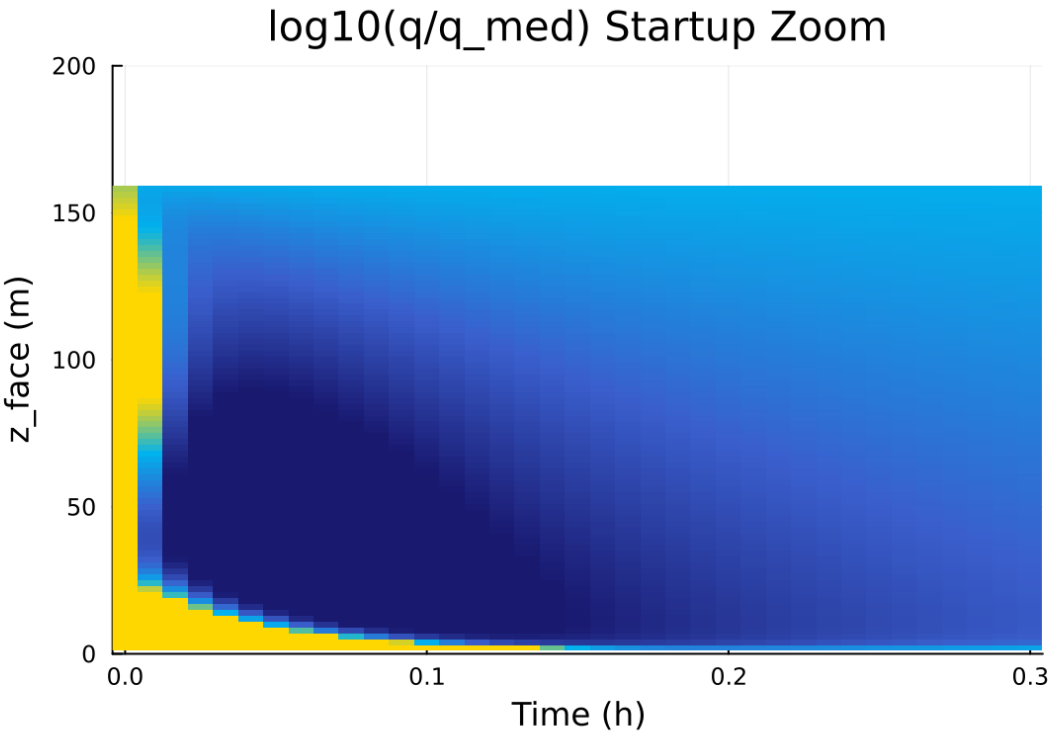

Figure 50: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

10 Case 5: gabls1 80x2m 9h report

11 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

11.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 102 / 0
- RHS evaluations: 25502
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

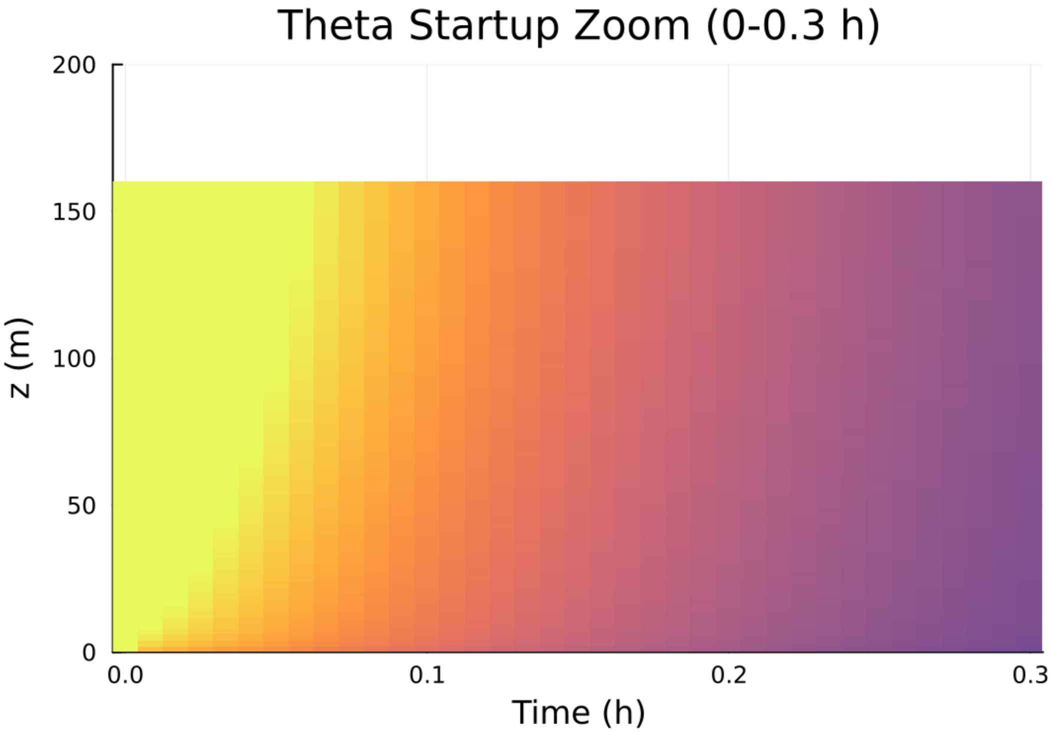

Figure 51: Fold proximity diagnostics versus time

Table 5: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$8.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	$0.100000\,\mathrm{m}$
Surface roughness (heat)	z_{0h}	$0.010000\,\mathrm{m}$
Coriolis parameter	f	$1.000\text{e-}04\,\mathrm{s^{-1}}$
Reference potential temperature	θ_a	$265.000000\,\mathrm{K}$
Deep soil temperature	T_{deep}	$262.000000\,\mathrm{K}$
Downward longwave forcing	$R_{\downarrow}$	$245.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	$0.100000\,\mathrm{m}$
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	$265.000000\,\mathrm{K}$
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

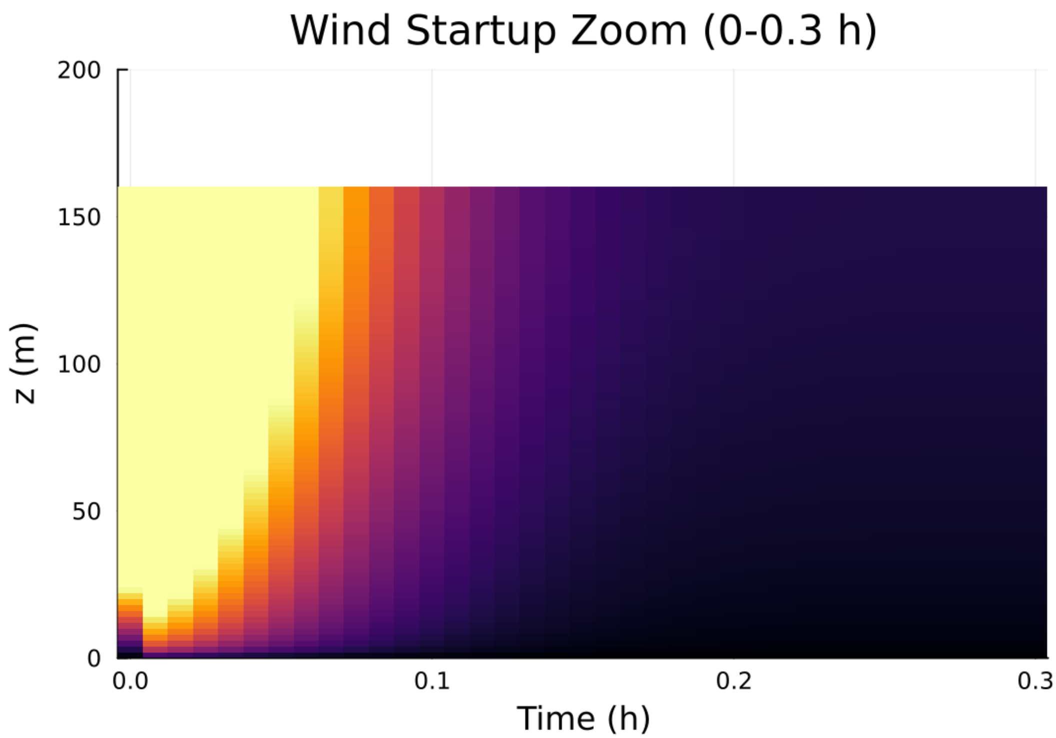

Figure 52: fig03c wind startup zoom.png

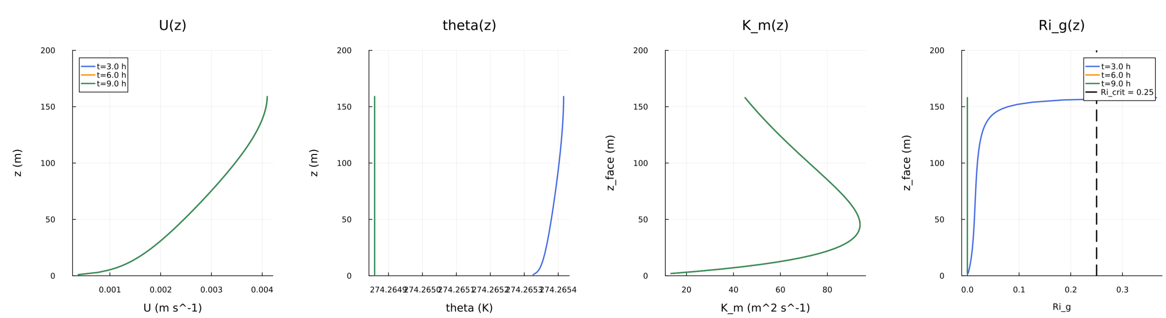

Figure 53: fig04 profiles u theta km.png

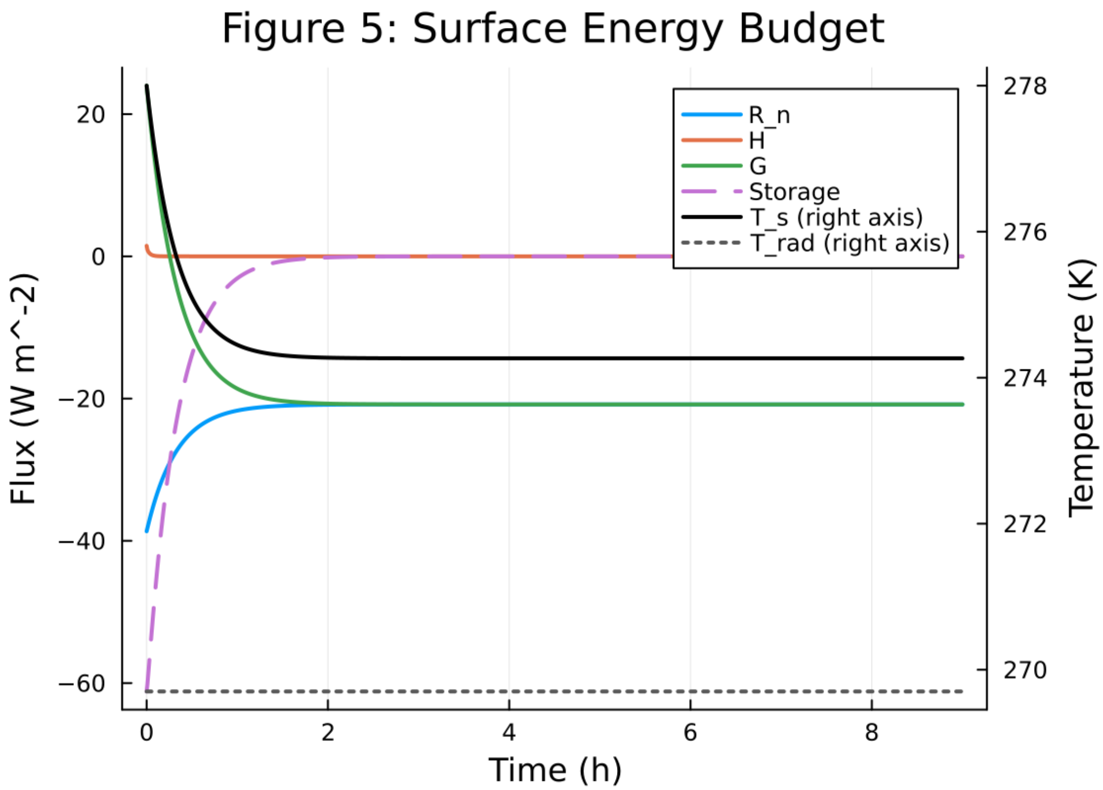

Figure 54: fig05 surface energy budget.png

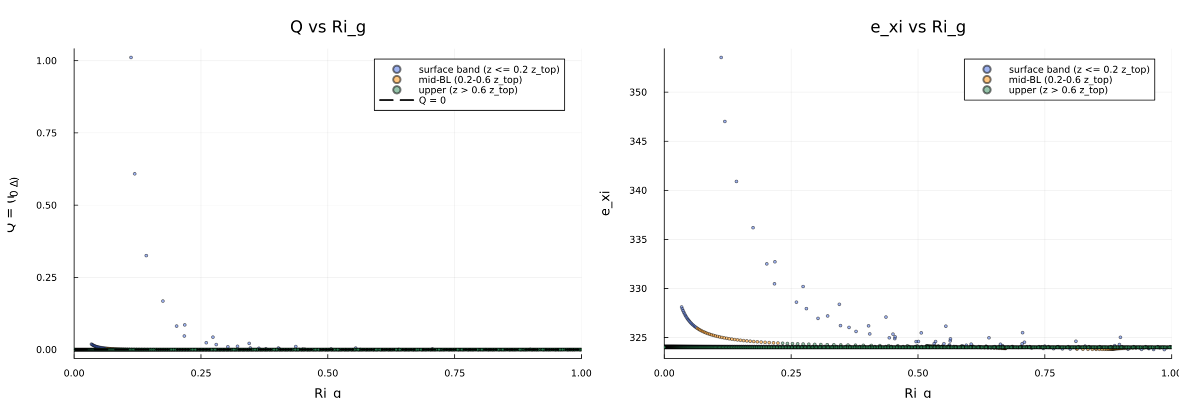

Figure 55: fig06 phase delta exi.png

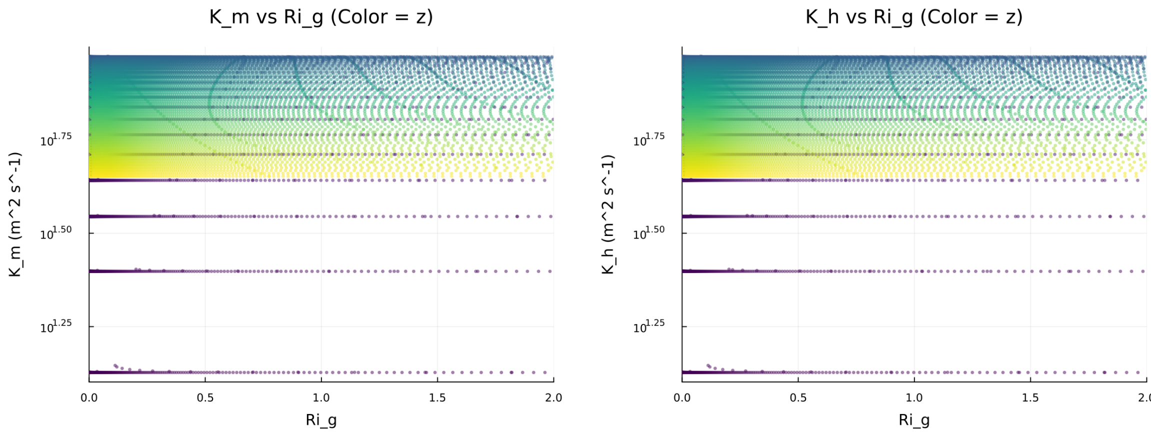

Figure 56: fig07 diffusivity vs ri.png

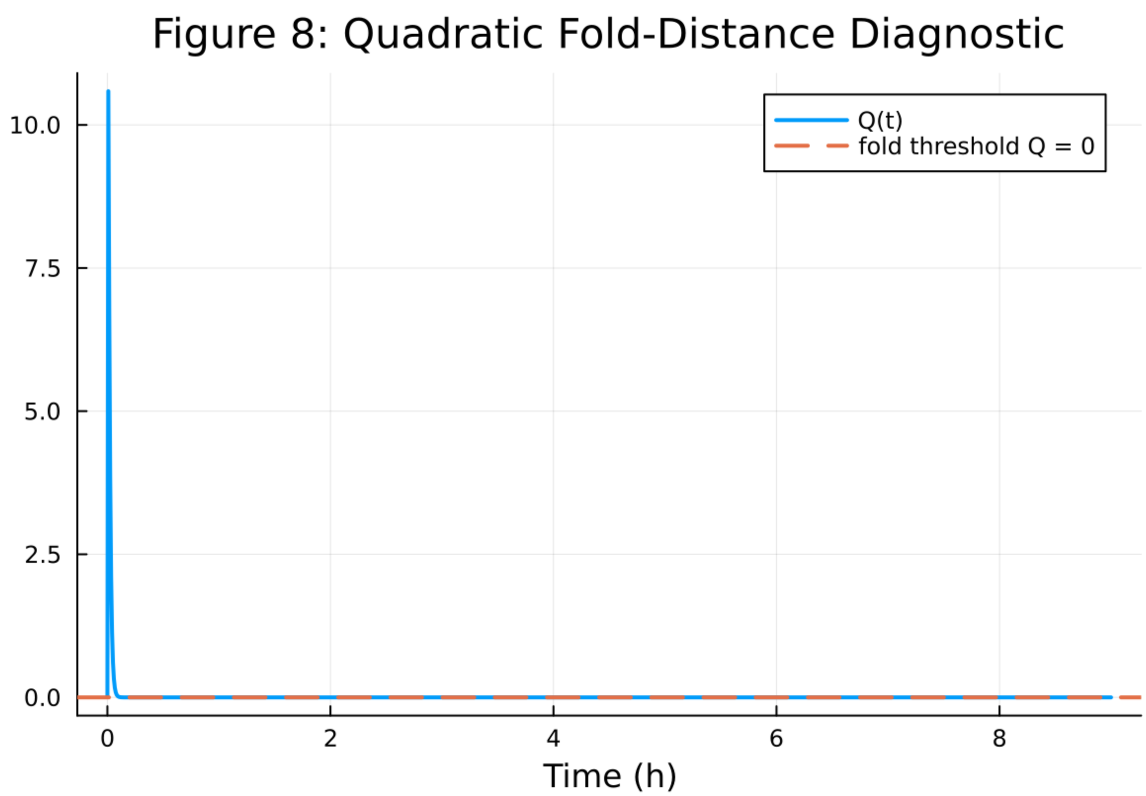

Figure 57: fig08 fold proximity.png

11.2 Forcing and Boundary Conditions

11.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 94.819611%
- Diffusivity bounds: $K_m \in [11.165526, 78.288180]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [-1.154 \times 10^{-6}, 1887.093112]$

11.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

11.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

11.6 Included Plots

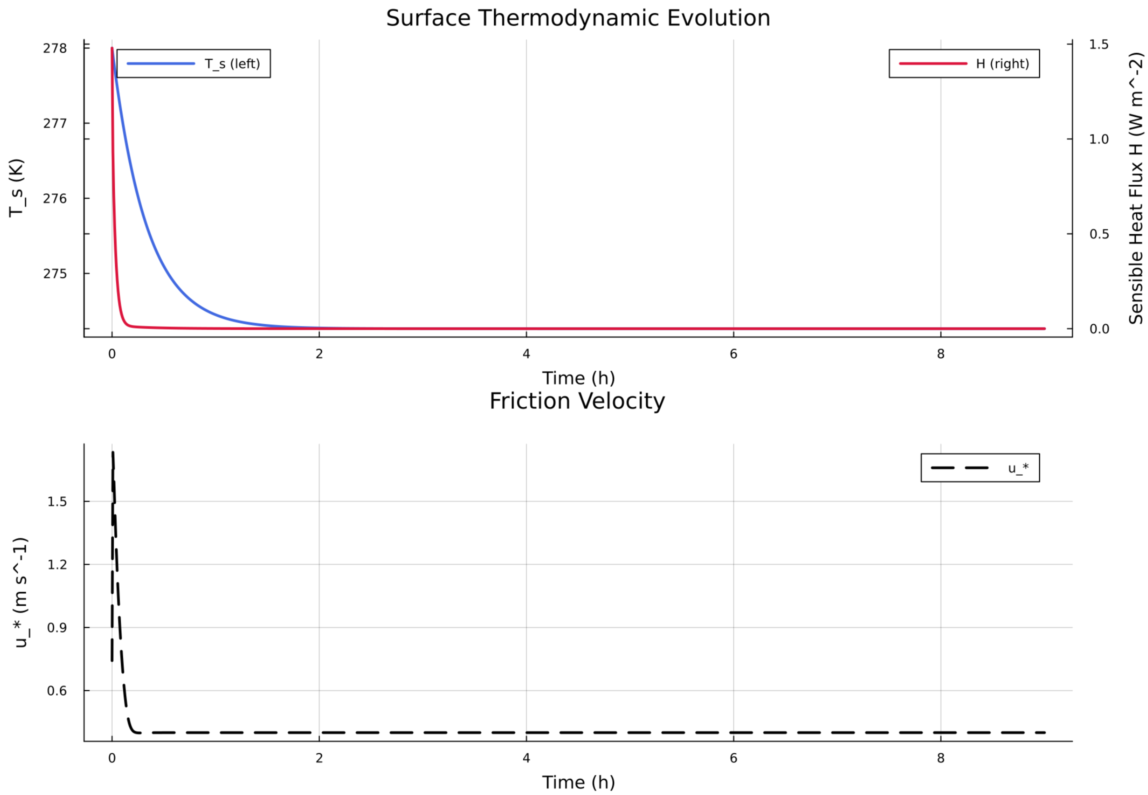

Figure 58: Time series of T__s, sensible heat flux, and friction velocity

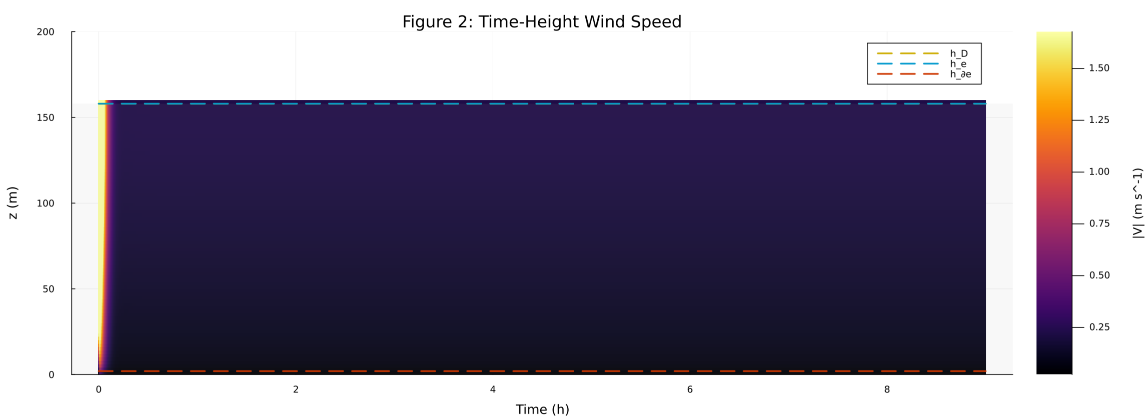

Figure 59: Time-height contour of wind speed with LLJ evolution

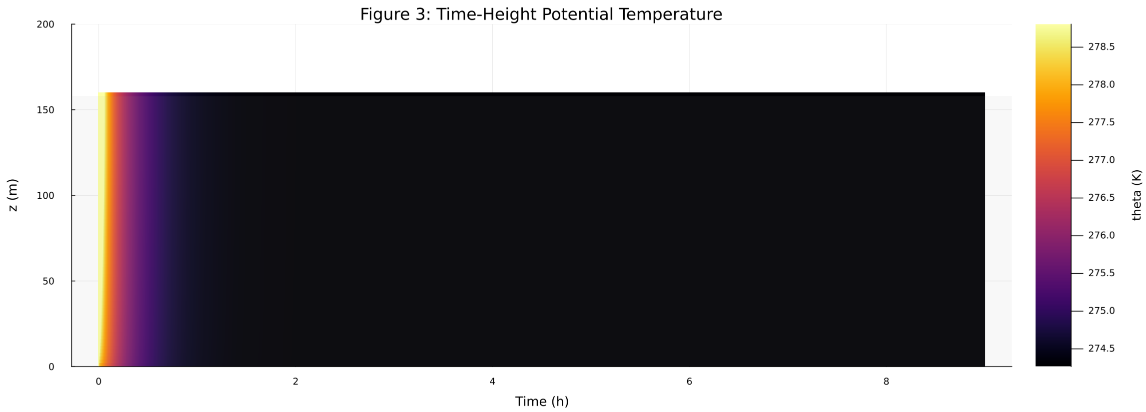

Figure 60: Time-height contour of potential temperature and inversion growth

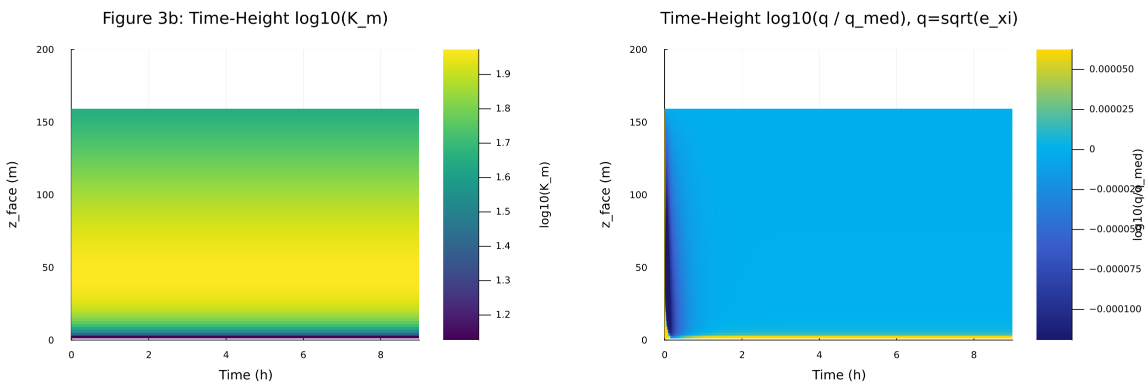

Figure 61: Vertical profiles of U, theta, and Km at representative times

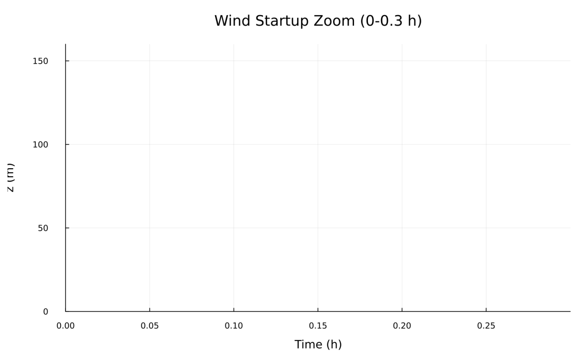

Figure 62: Surface energy budget components (R_n, H, G, storage)

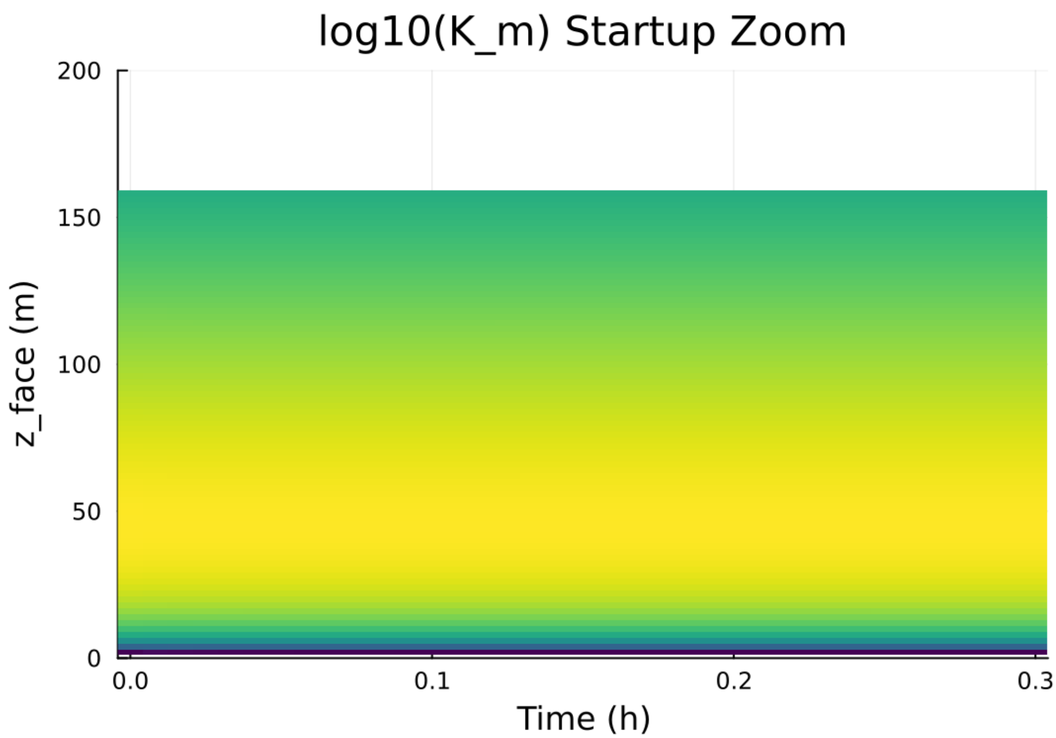

Figure 63: Phase portrait on regularized manifold (Delta vs e_xi)

12 Case 6: idealized sbl 80x2m 9h report

13 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

13.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

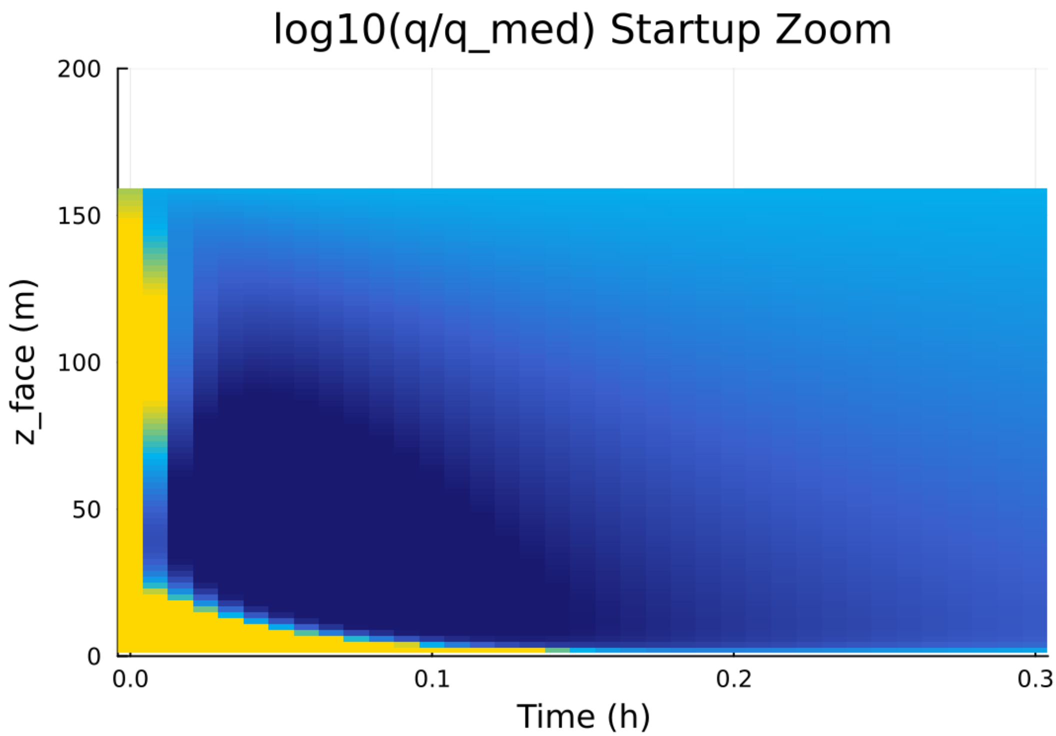

Figure 64: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

Table 6: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$10.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	$0.020000\,\mathrm{m}$
Surface roughness (heat)	z_{0h}	$0.005000\,\mathrm{m}$
Coriolis parameter	f	$1.000\text{e-}04\,\mathrm{s^{-1}}$
Reference potential temperature	θ_a	$280.000000\,\mathrm{K}$
Deep soil temperature	T_{deep}	$276.000000\,\mathrm{K}$
Downward longwave forcing	$R_{\downarrow}$	$300.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	$0.100000\,\mathrm{m}$
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	$265.000000\,\mathrm{K}$
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

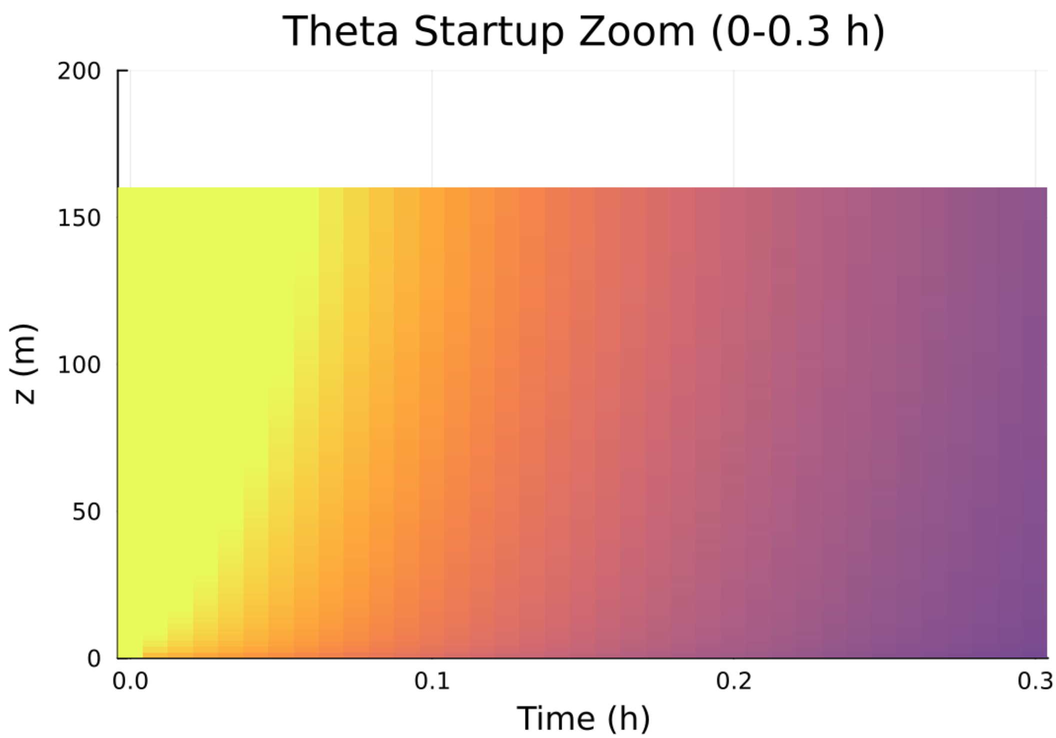

Figure 65: Fold proximity diagnostics versus time

13.2 Forcing and Boundary Conditions

13.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399989, 94.108839]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [-4.024 \times 10^{-6}, 1551.592844]$

13.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

13.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

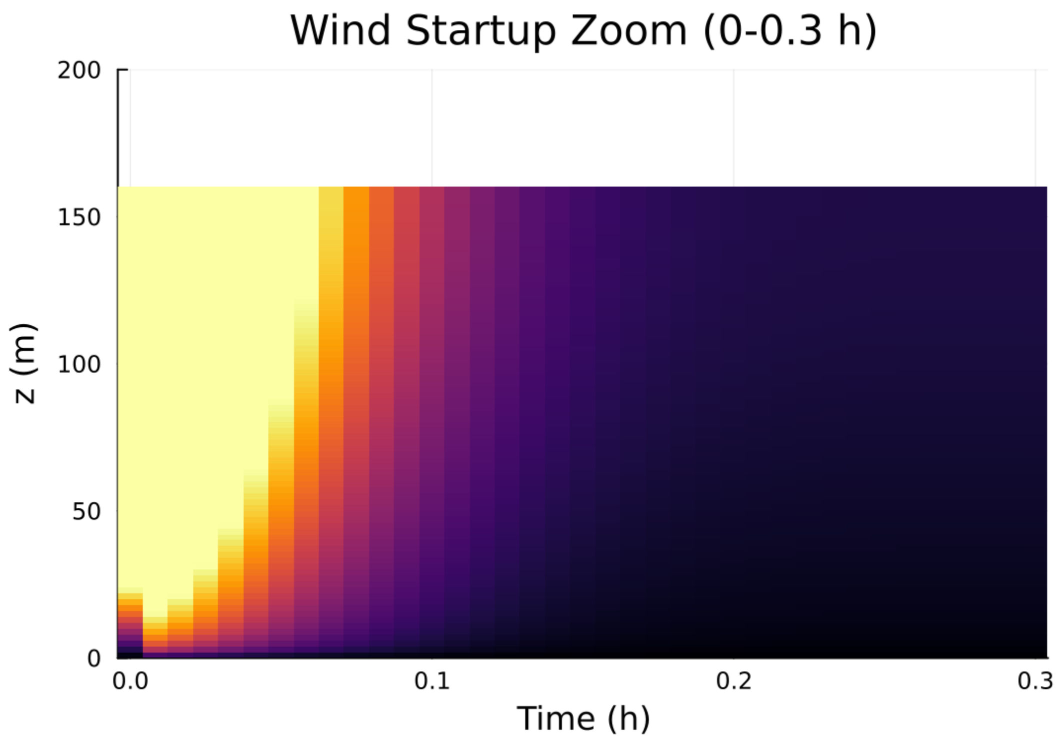

Figure 66: fig03c wind startup zoom.png

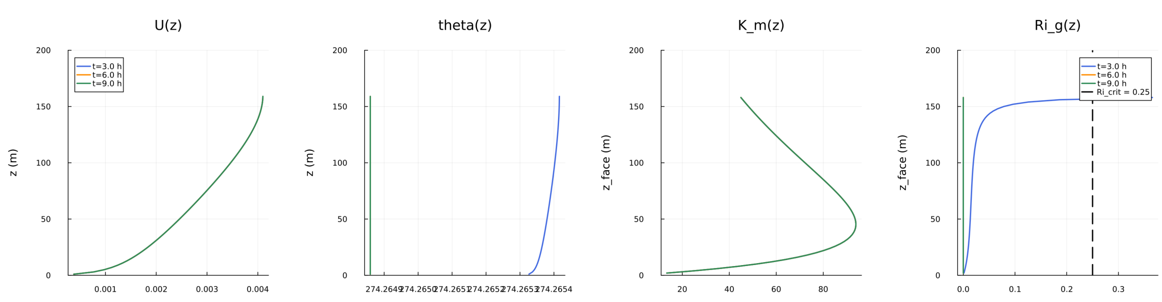

Figure 67: fig04 profiles u theta km.png

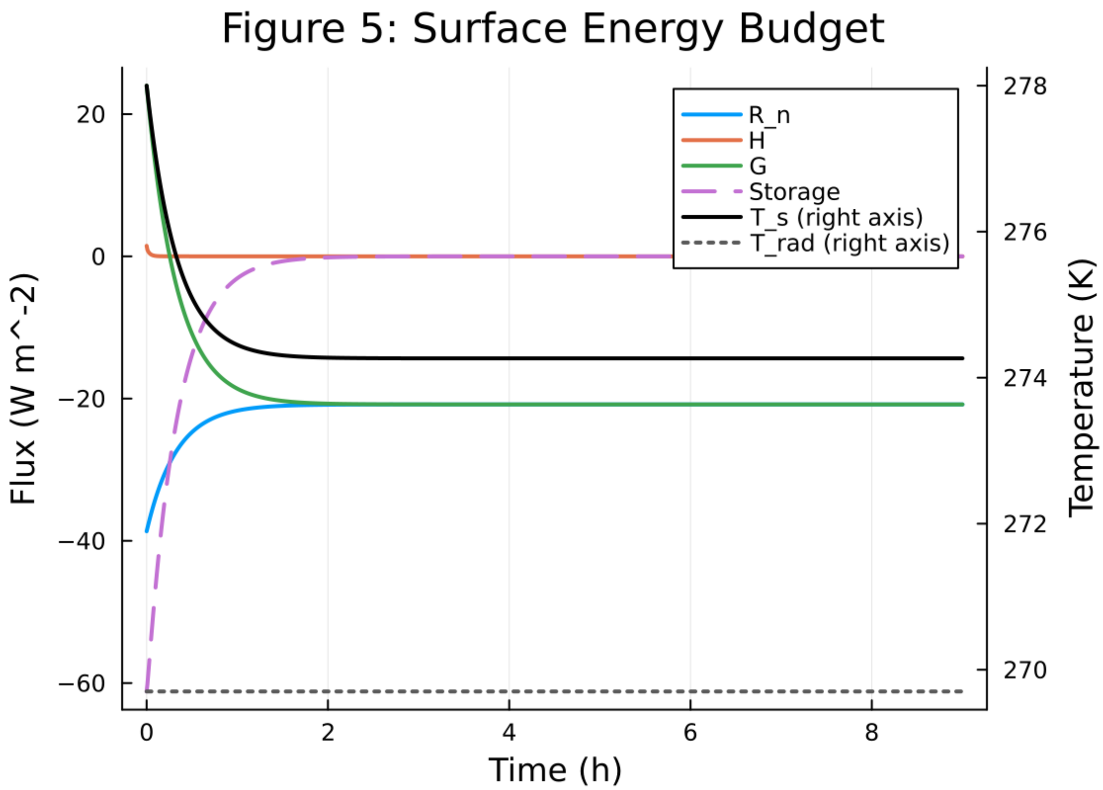

Figure 68: fig05 surface energy budget.png

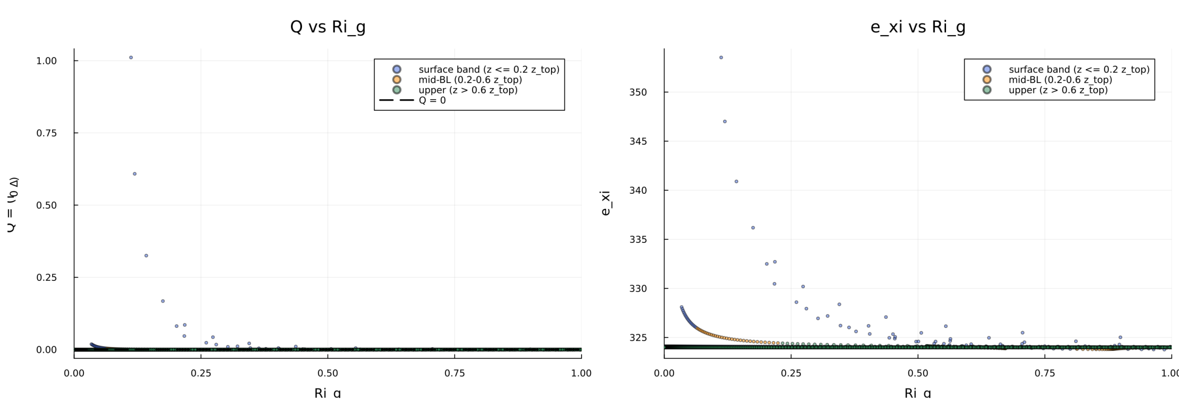

Figure 69: fig06 phase delta exi.png

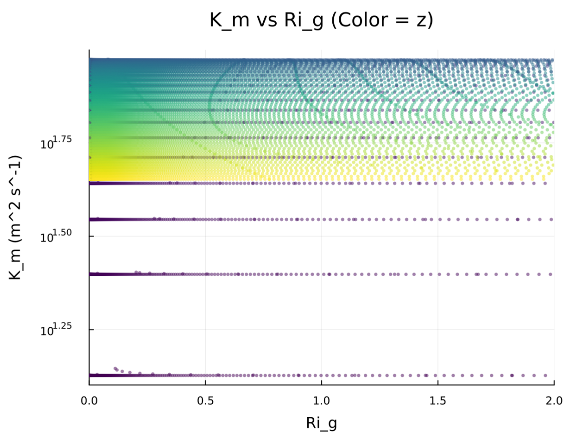

Figure 70: fig07 diffusivity vs ri.png

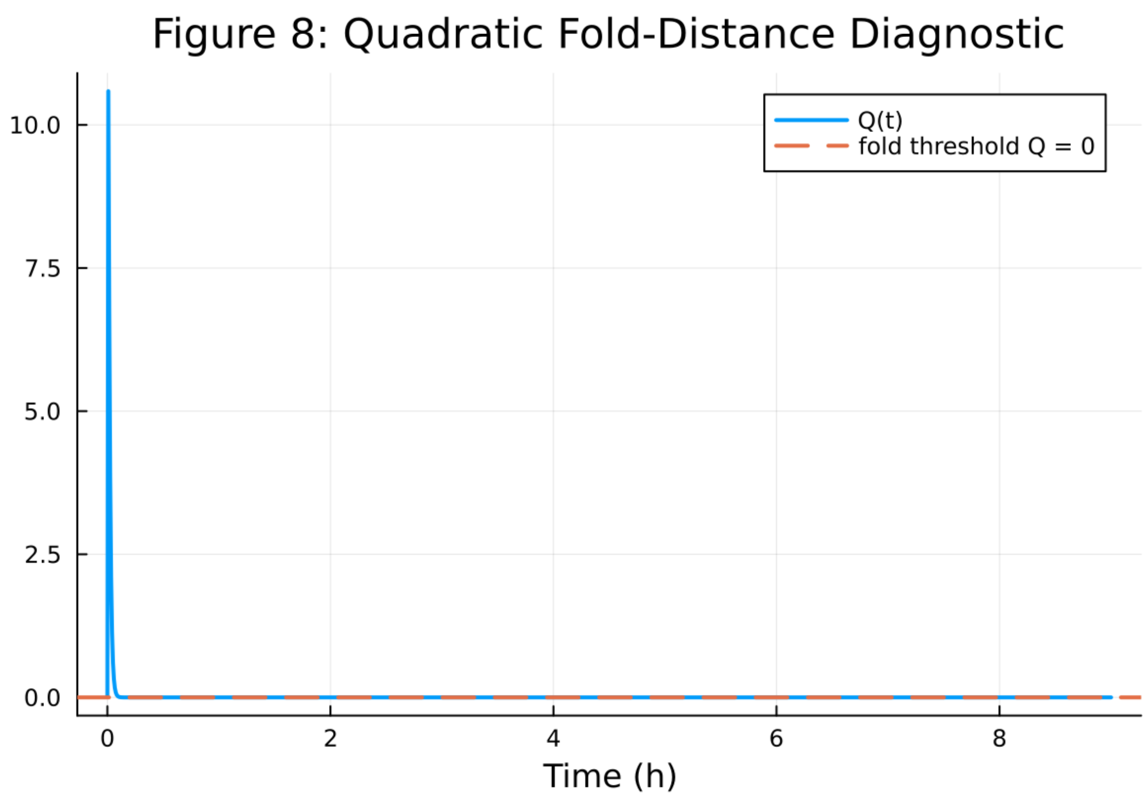

Figure 71: fig08 fold proximity.png

5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

13.6 Included Plots

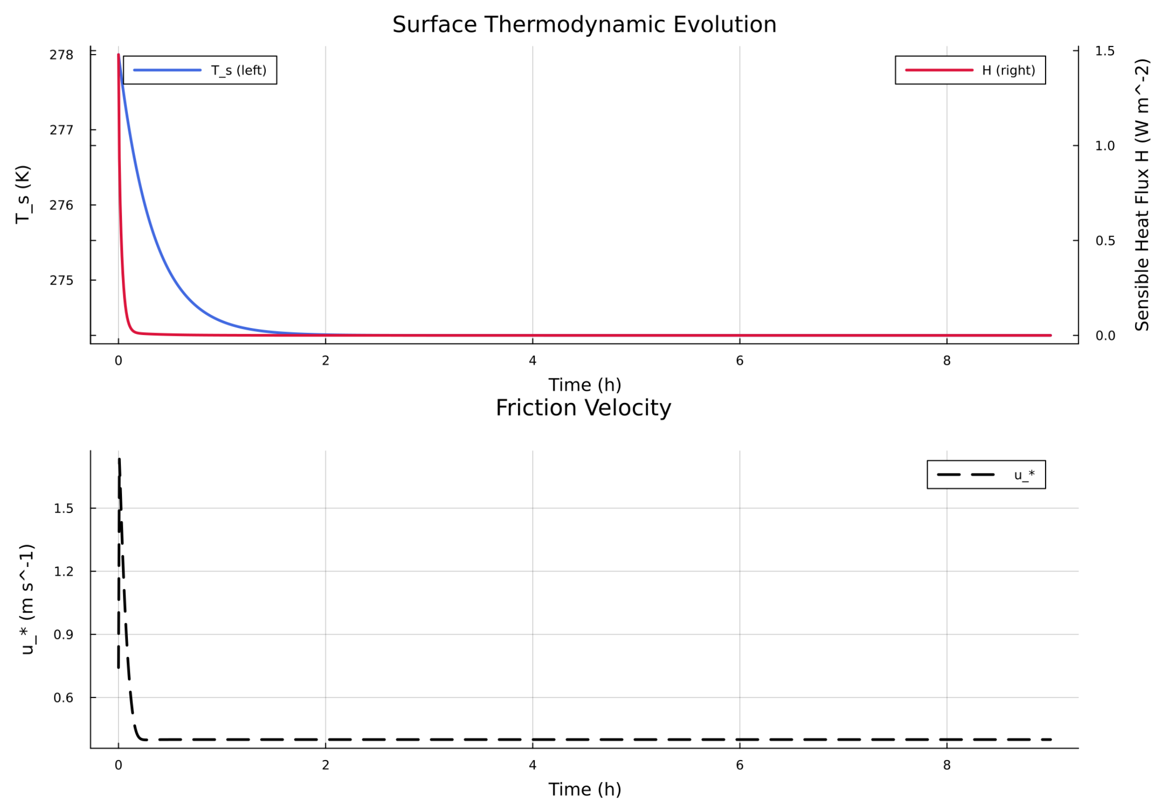

Figure 72: Time series of T__s, sensible heat flux, and friction velocity

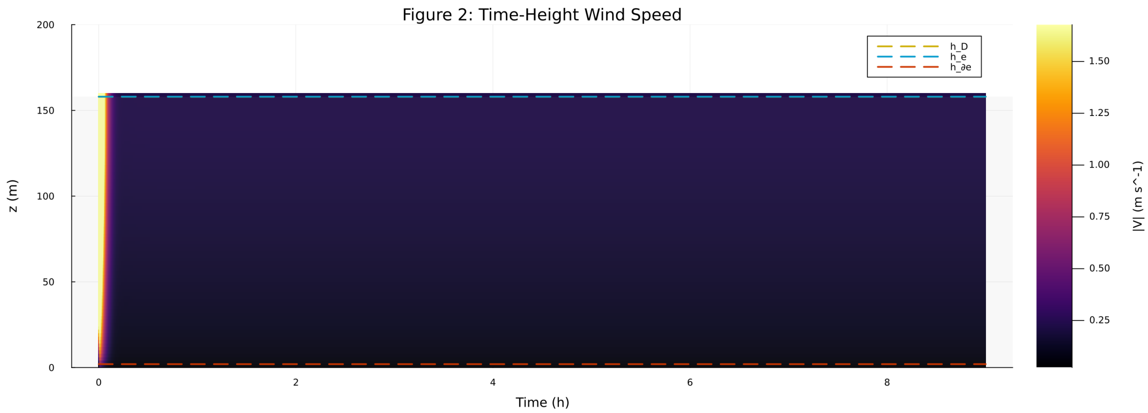

Figure 73: Time-height contour of wind speed with LLJ evolution

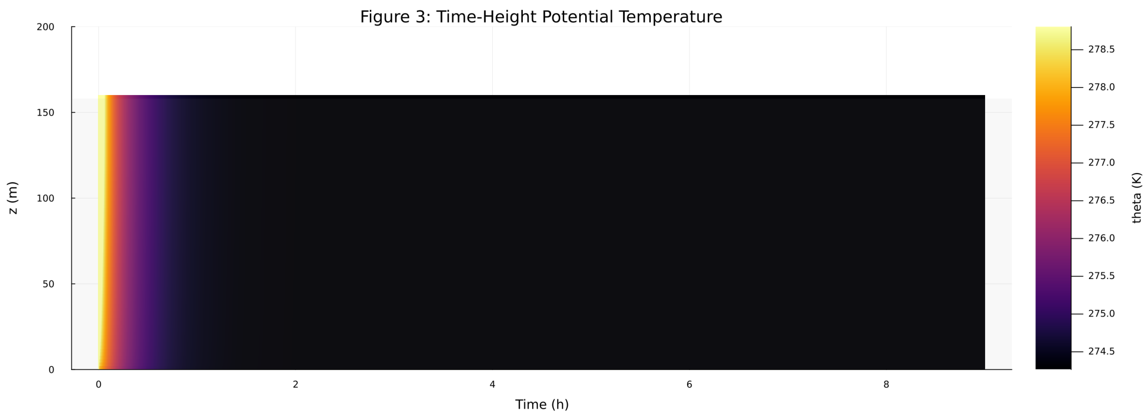

Figure 74: Time-height contour of potential temperature and inversion growth

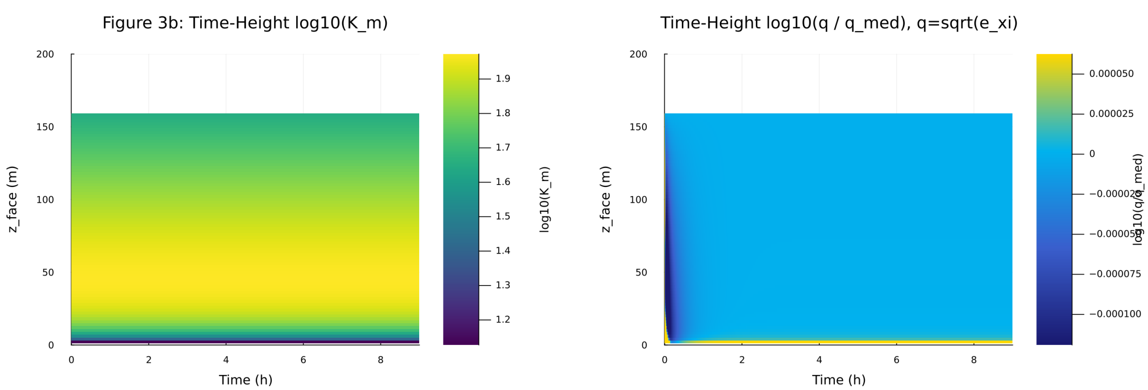

Figure 75: Vertical profiles of U, theta, and Km at representative times

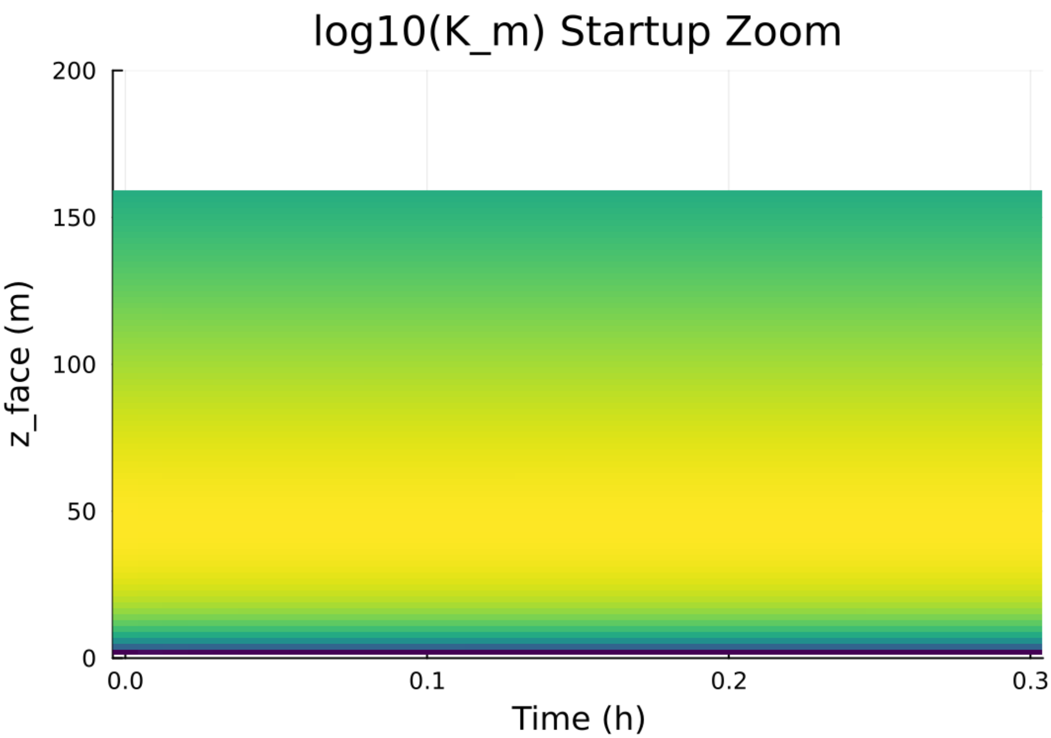

Figure 76: Surface energy budget components (R_n, H, G, storage)

14 Case 7: scm verify 40x4m 0h report

15 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/scm_verify/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

15.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/scm_verify`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 52 / 0
- RHS evaluations: 6762
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

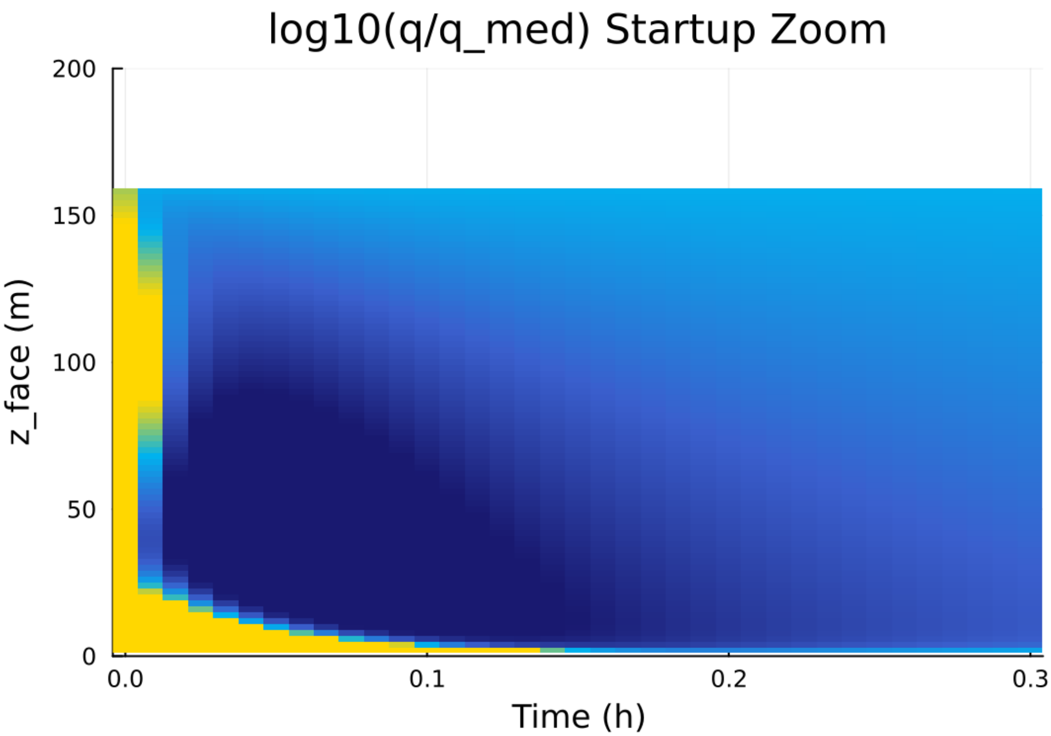

Figure 77: Phase portrait on regularized manifold (Delta vs e_xi)

Table 7: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$10.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	$0.020000\,\mathrm{m}$
Surface roughness (heat)	z_{0h}	$0.005000\,\mathrm{m}$
Coriolis parameter	f	$1.000\text{e-}04\,\mathrm{s^{-1}}$
Reference potential temperature	θ_a	$280.000000\,\mathrm{K}$
Deep soil temperature	T_{deep}	$276.000000\,\mathrm{K}$
Downward longwave forcing	$R_{\downarrow}$	$300.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	$0.100000\,\mathrm{m}$
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	$265.000000\,\mathrm{K}$
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

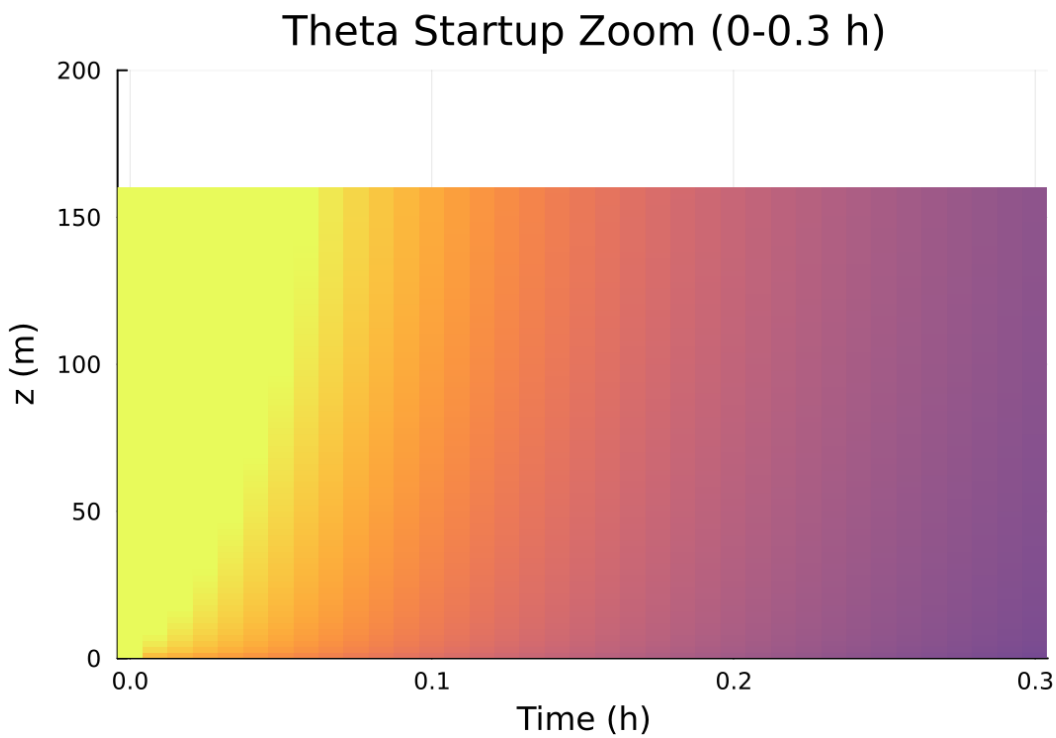

Figure 78: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

15.2 Forcing and Boundary Conditions

15.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [25.002388, 94.099429]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.035577, 103.545808]$

15.4 Generated Artifacts

- Time-series CSV: `results/scm_verify/time_series.csv`
- Diagnostic summary JSON: `results/scm_verify/summary.json`
- Payload JLD2: `results/scm_verify/payload.jld2`

15.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

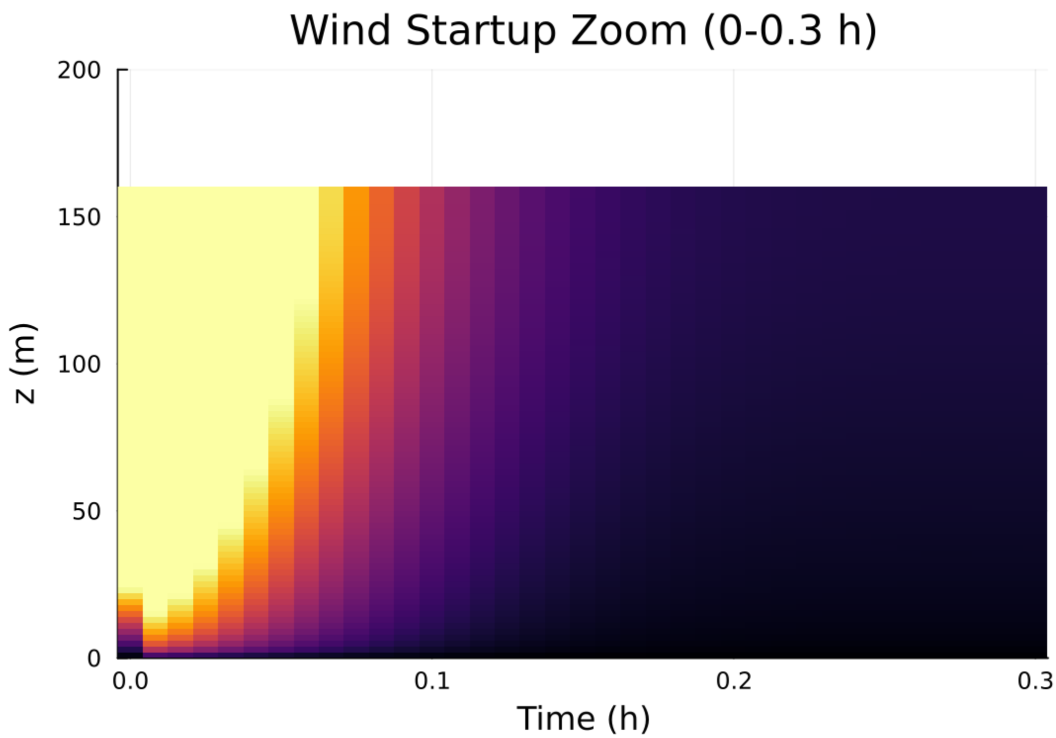

Figure 79: Fold proximity diagnostics versus time

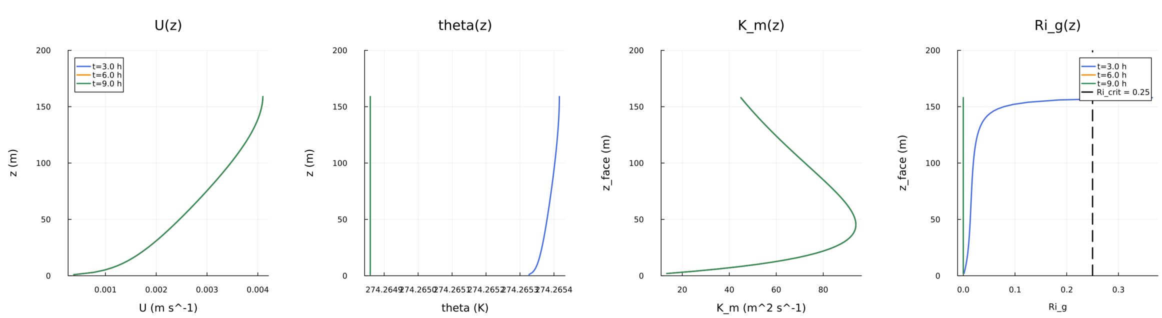

Figure 80: fig04 profiles u theta km.png

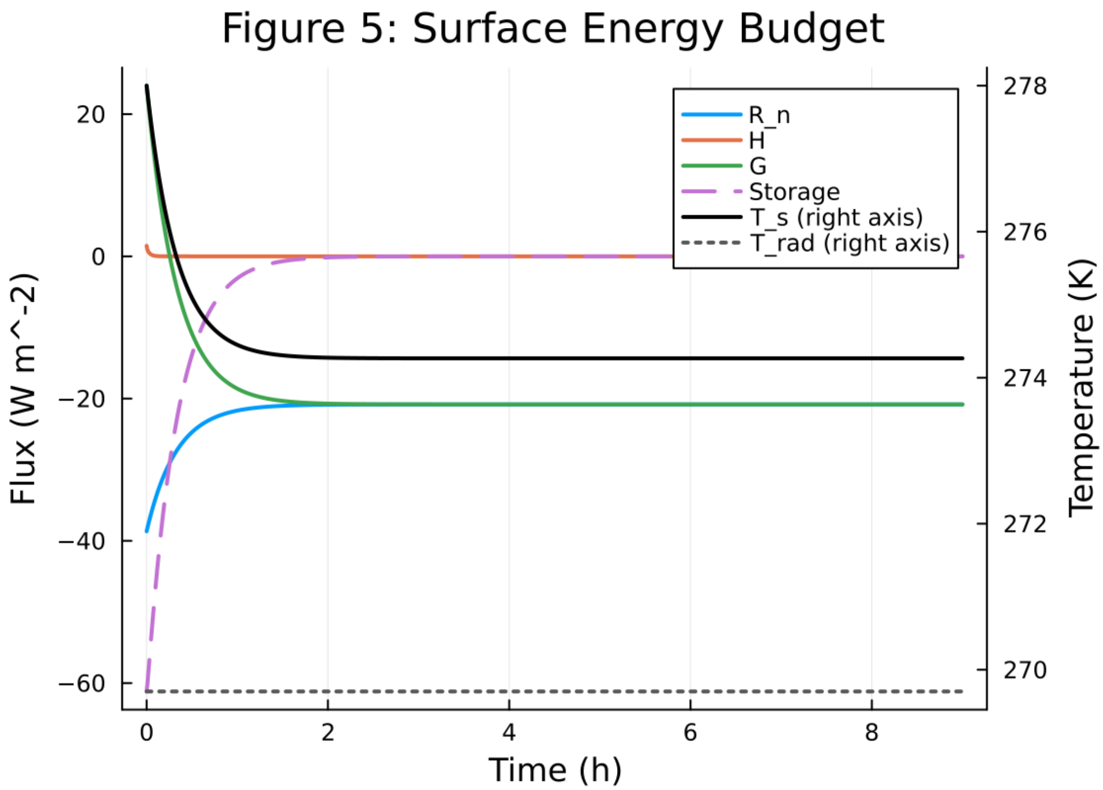

Figure 81: fig05 surface energy budget.png

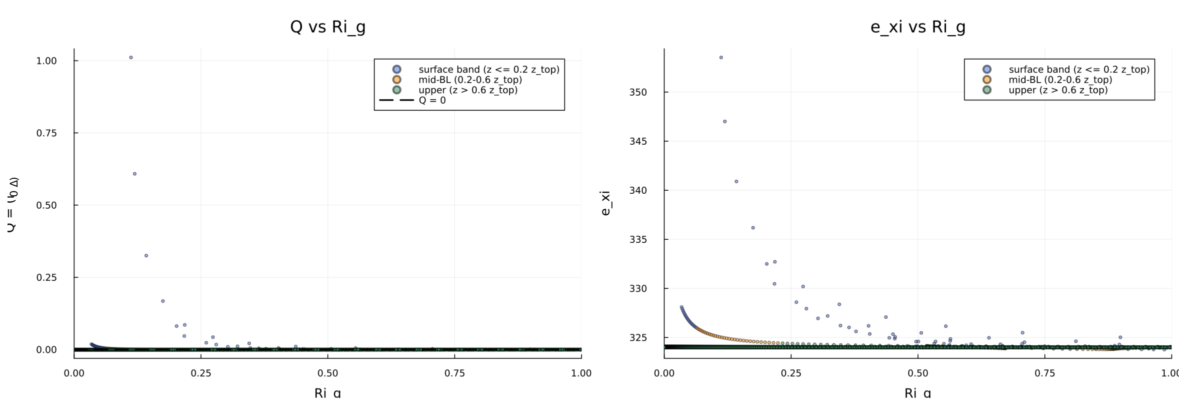

Figure 82: fig06 phase delta exi.png

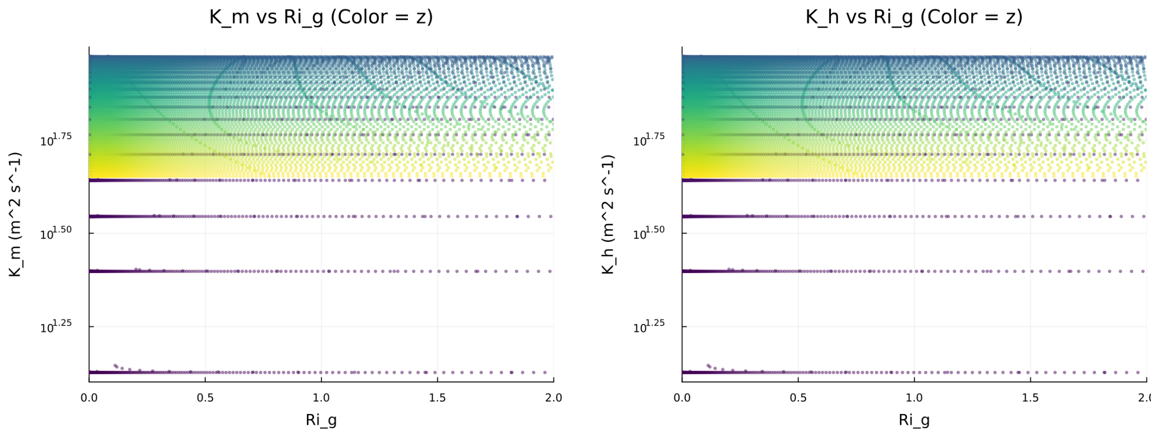

Figure 83: fig07 diffusivity vs ri.png

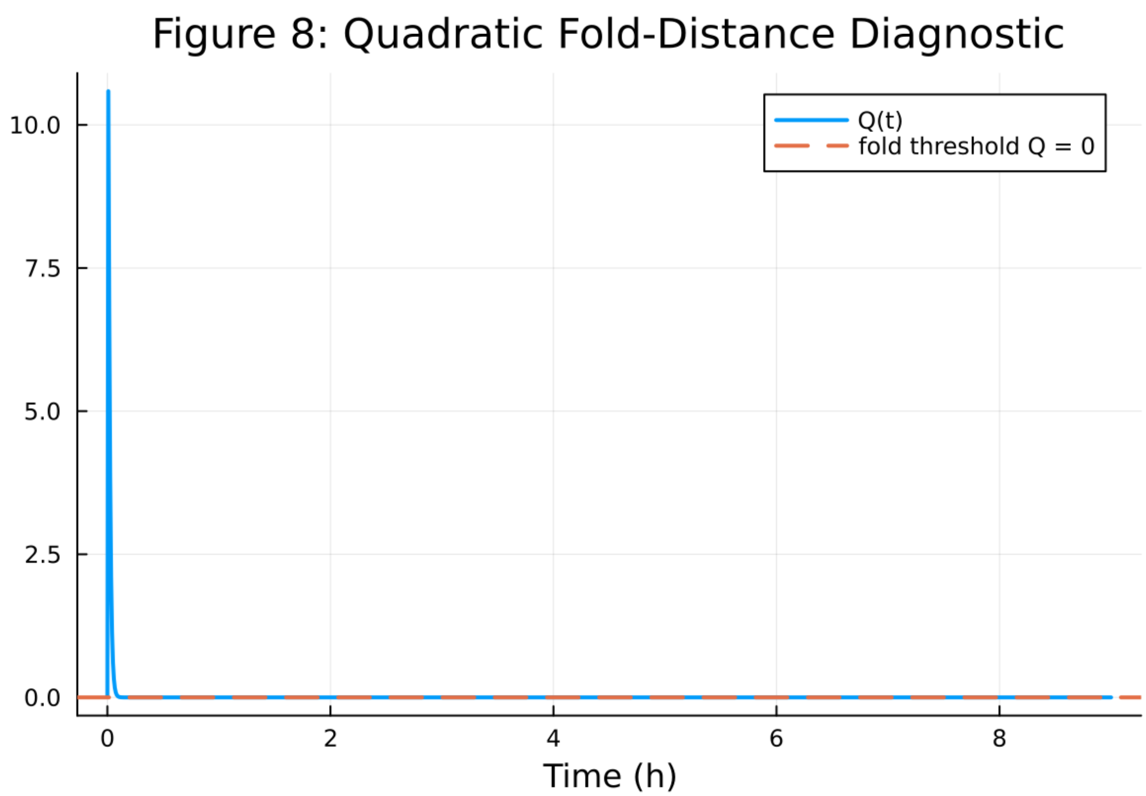

Figure 84: fig08 fold proximity.png

5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

15.6 Included Plots

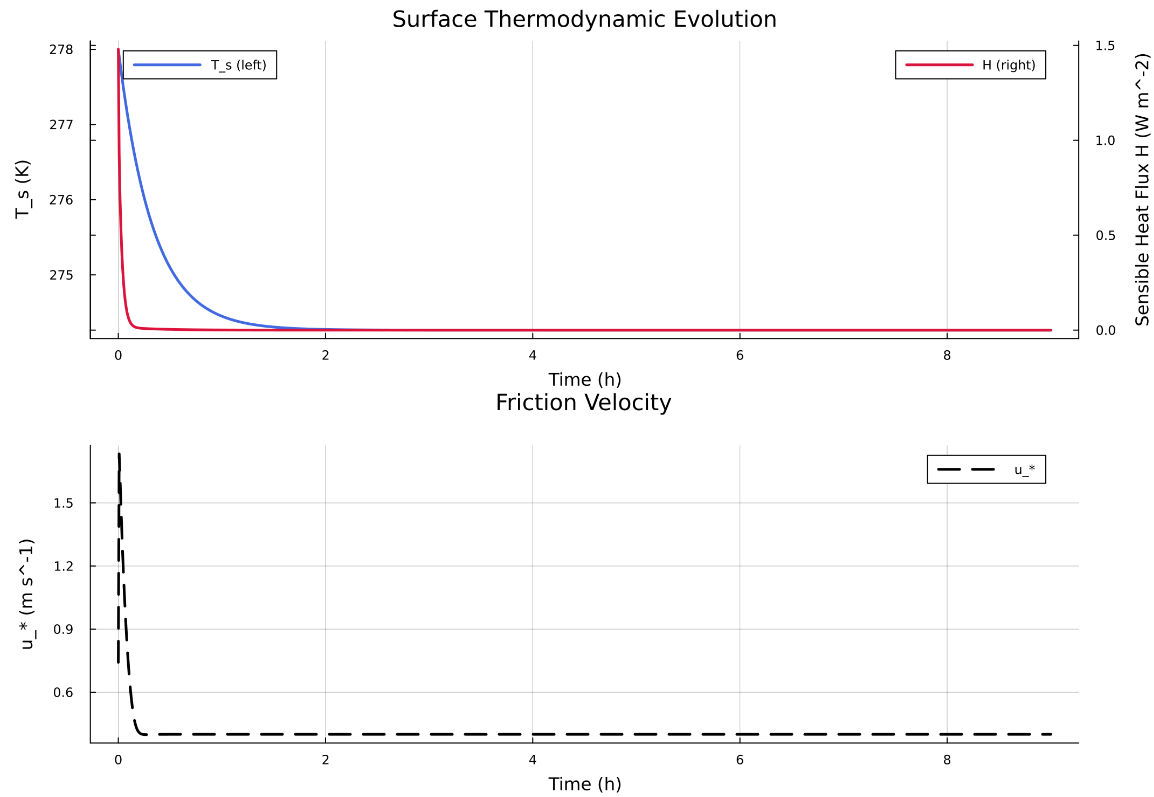

Figure 85: Time series of T__s, sensible heat flux, and friction velocity

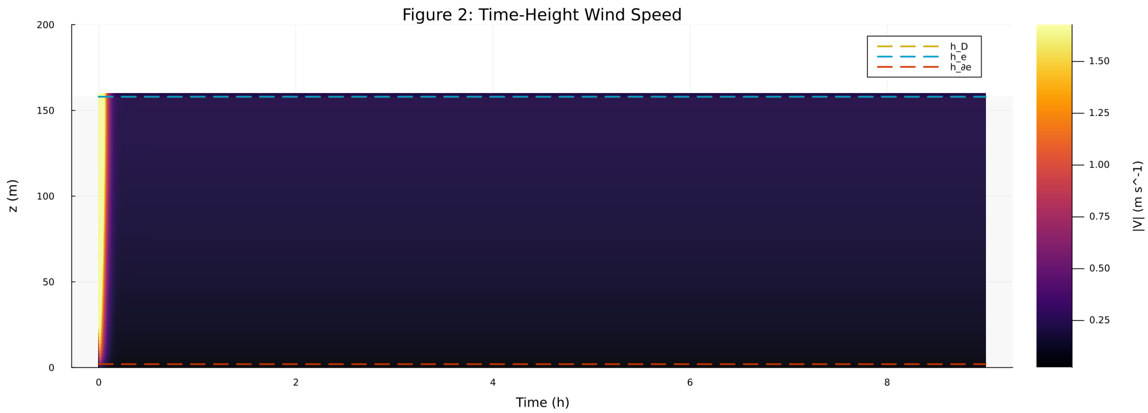

Figure 86: Time-height contour of wind speed with LLJ evolution

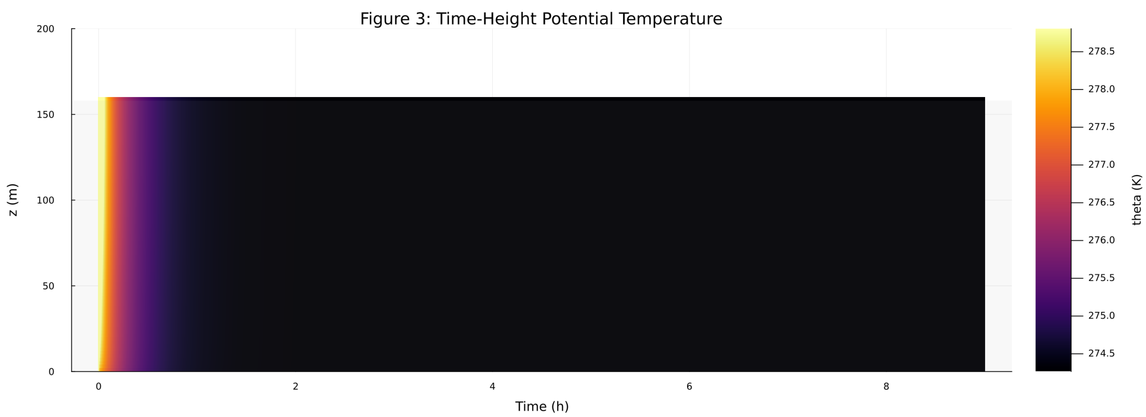

Figure 87: Time-height contour of potential temperature and inversion growth

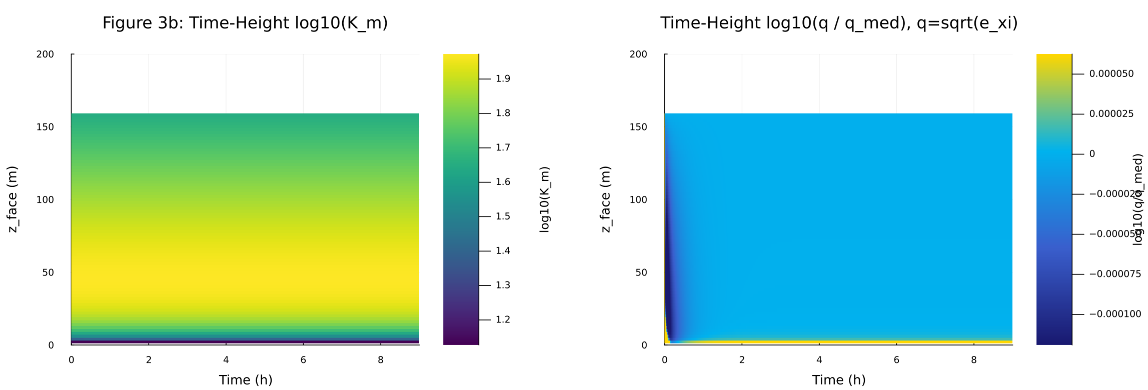

Figure 88: Vertical profiles of U, theta, and Km at representative times

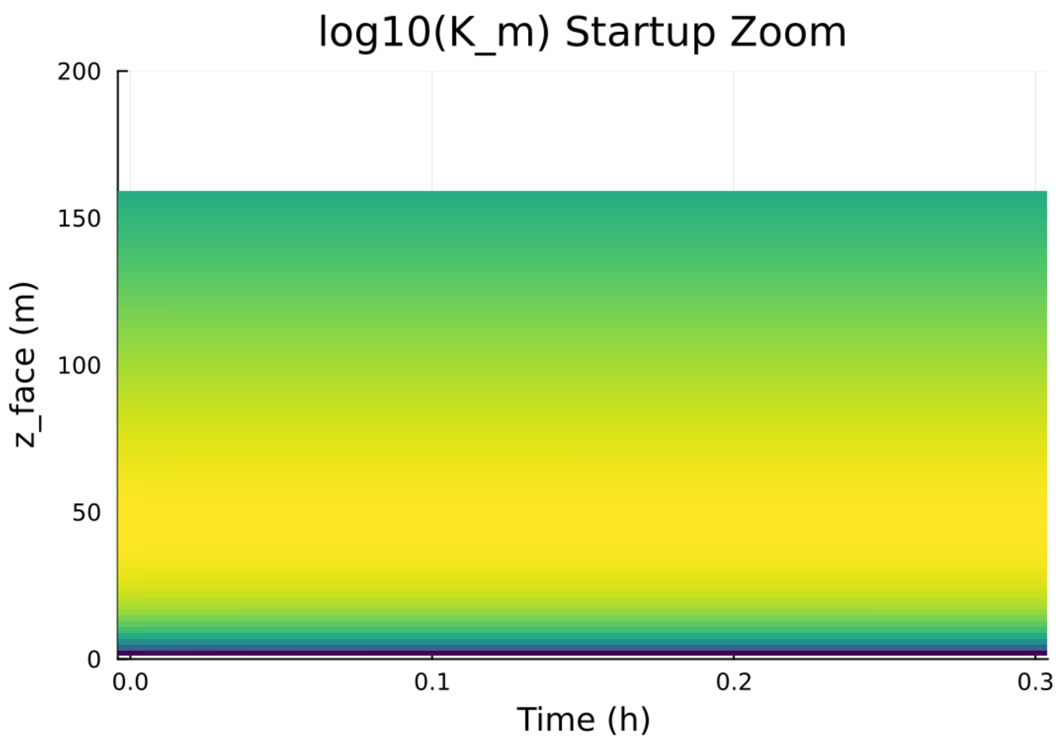

Figure 89: Surface energy budget components (R_n, H, G, storage)

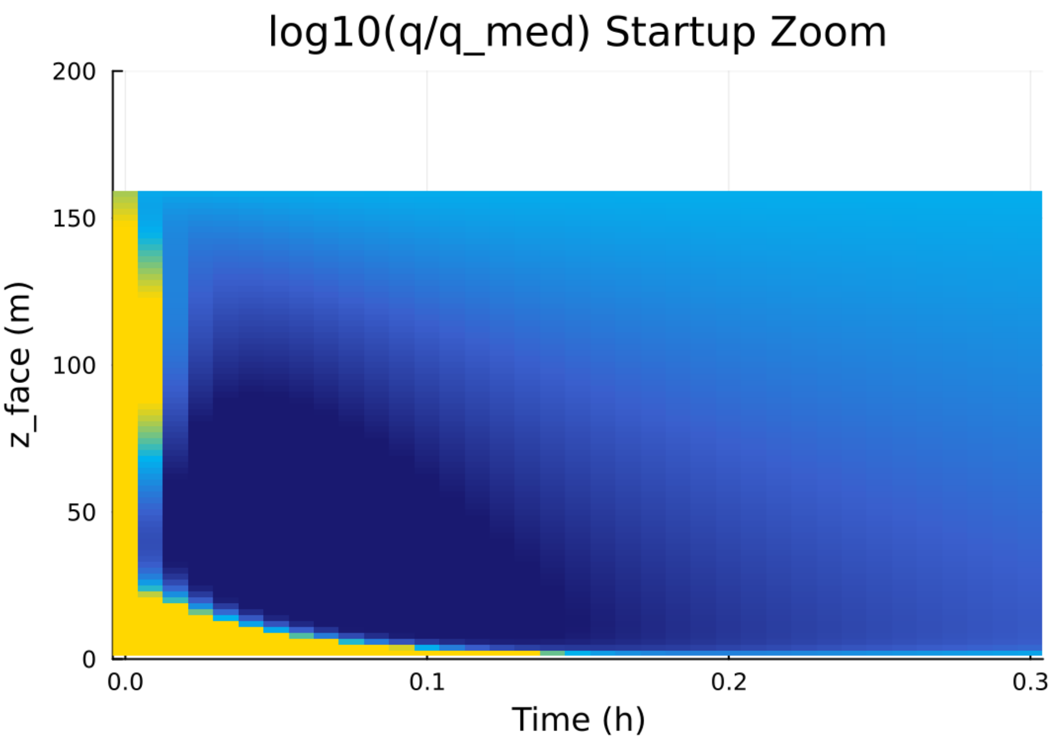

Figure 90: Phase portrait on regularized manifold (Delta vs e__xi)

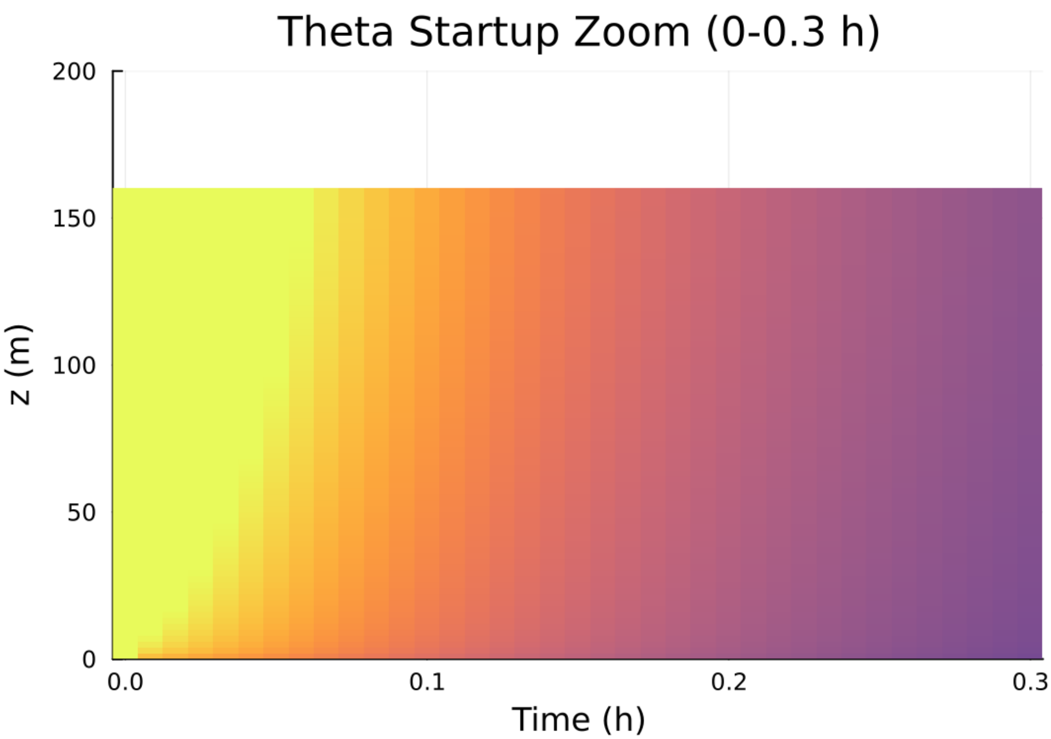

Figure 91: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

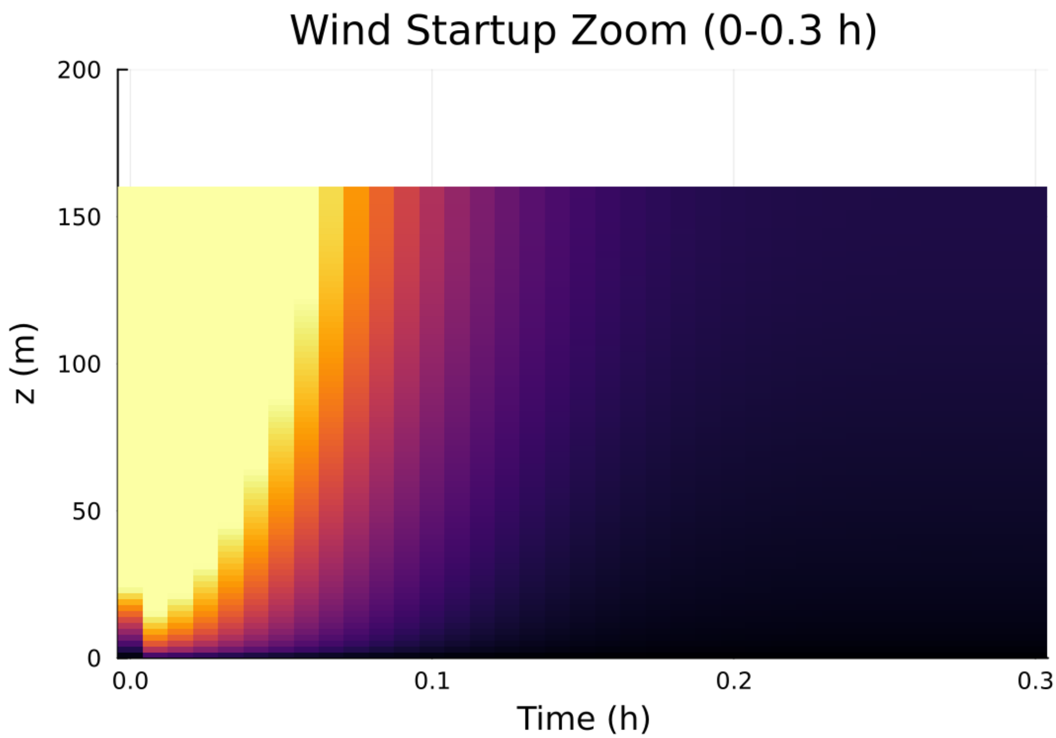

Figure 92: Fold proximity diagnostics versus time

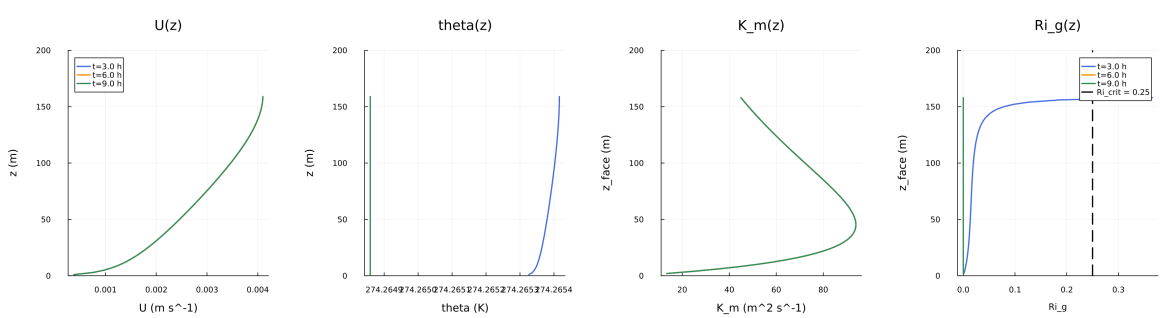

Figure 93: fig04 profiles u theta km.png

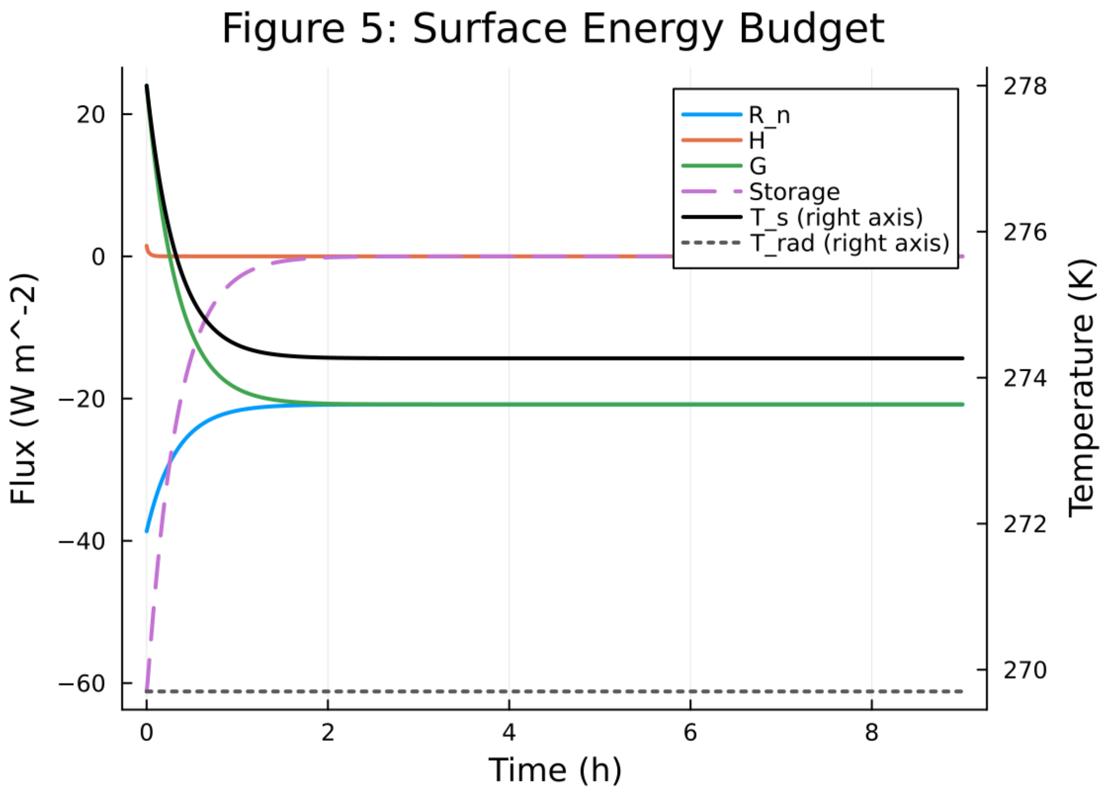

Figure 94: fig05 surface energy budget.png

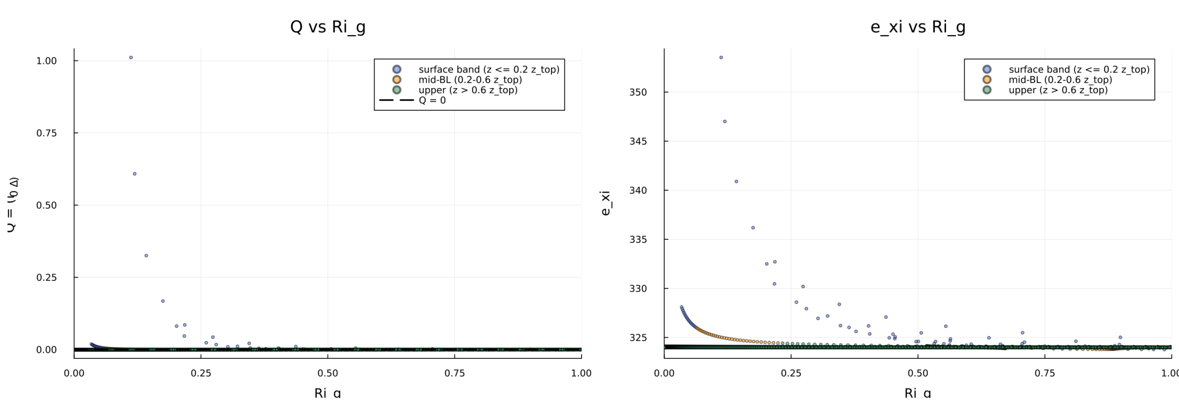

Figure 95: fig06 phase delta exi.png

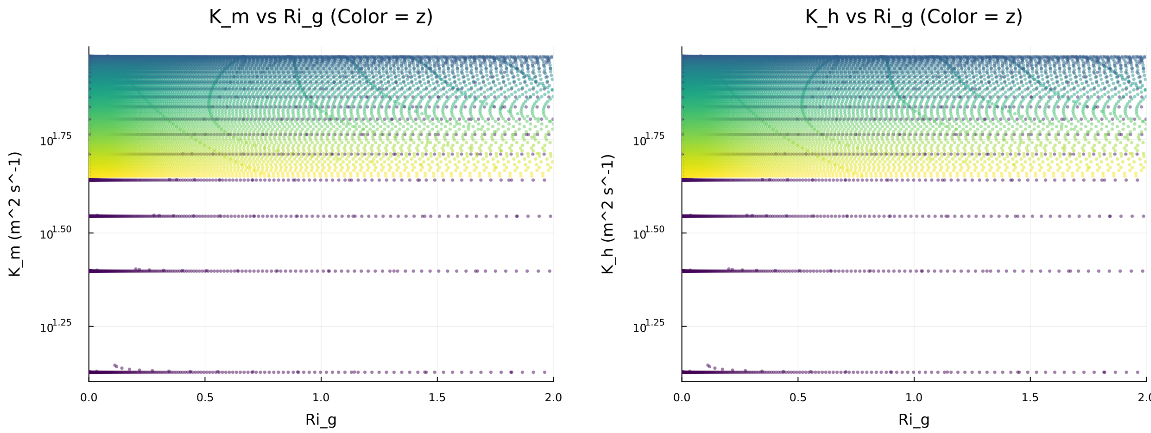

Figure 96: fig07 diffusivity vs ri.png

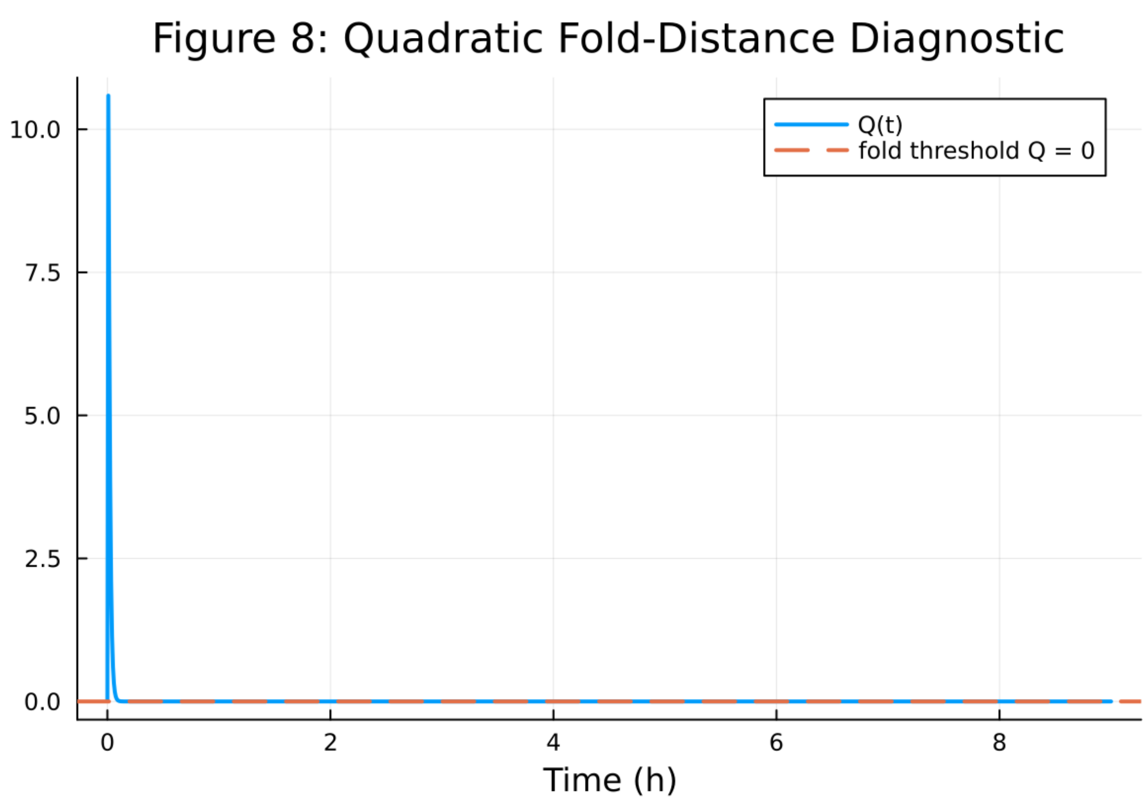

Figure 97: fig08 fold proximity.png